How much are candidates willing to pay for more openings and higher chances of getting the job? Evidence from Civil Service Exams *

Deborah Doukas †

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Abstract

I document that job seekers have intrinsic preferences over the competitiveness of getting the job positions they apply for in a high-stakes, real-world setting. I do this by leveraging a unique natural setting in Brazil where job applicants apply for government positions using a Deferred Acceptance Algorithm. Because the algorithm is incentive compatible, applicants' rankings of positions should reflect the positions' competitiveness only through their true preferences. I estimate these preferences using two measures: the probability that each candidate is matched with a certain job and the number of openings available for a particular position. My estimates imply that high-performing candidates are willing to pay, on average, 3.6 to 5.7 percent of the earnings provided by a job posting to increase their matching probability by 10 percentage points and 0.2 to 2.3 percent to have 10 additional openings in a job posting. These results suggest that application procedures reveal a desire to avoid competitive job positions not only due to application costs, but also because individuals have direct preferences over the difficulty of obtaining a job.

Keywords: Competition, Job Search, Non-Monetary Job Attributes, Public Sector Labor Markets
JEL Codes: D4, J32, J45, M5

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†Department of Economics, Cornell University

1 Introduction

Understanding how people value non-pecuniary aspects of jobs is important for comprehending how labor markets function. Existing evidence indicates that some non-pecuniary aspects influence individuals' decisions by signaling how competitive a job is to get. On online job platforms, information about the number of applicants for a specific position or the age of a job posting could affect candidates' willingness to apply for those vacancies (Gee (2019), Fradkin et al. (2025)). However, evidence on how competition shapes candidates' interaction with application procedures outside of an experimental setting is still scarce. This is particularly relevant in the context of major emerging economies such as Brazil, China, and India, where millions of candidates face extremely competitive civil service exams to apply for government positions.

In this paper, I document that job candidates have intrinsic preferences over the competitiveness of getting the job positions they apply for in a high-stakes, real-world setting. There are not many measures of competitiveness explored in the literature that can be administered outside of a laboratory environment (Buser et al. (2026)). When researchers cannot directly elicit the subjective measures of competitiveness that candidates consider when making their application decisions, it is difficult to identify the effects of competitiveness on labor market outcomes. I analyze two measures that can be inferred simultaneously from the application procedure and identify how candidates respond to each measure individually. In addition, rather than focusing on the decision to apply to a job posting, this project examines how competitiveness influences candidates' rankings of alternatives to which they choose to apply, leveraging a unique natural setting in Brazil.

For a long time, a large fraction of job candidates seeking positions in the Brazilian public sector had to take one exam for each job posting they applied for, which was very costly, both for them and for the government. In 2024, the Brazilian government launched the first edition of the *Concurso Público Nacional Unificado* (CPNU), a new program in which job applicants took only one exam and were allowed to apply for many government positions. After taking the exam, the candidates submitted their ranked-ordered lists of positions through an online platform, and were matched to the job postings through a candidate-proposing Deferred Acceptance Algorithm. Because the algorithm is an incentive compatible centralized matching mechanism (Dubins and Freedman (1981), Roth (1982)), applicants' rankings should reflect only their true preferences over the job postings' attributes relevant to the application procedure.

The competitiveness associated with getting matched to a position is the main attribute of interest in this paper. I elicit competitiveness through two measures: the number of openings available in each job posting and the probability that each candidate is matched to a certain job, regardless of how they ranked it. I extracted the first measure directly from the job descriptions and it was observable by the candidates before they submitted their ranked preferences. The second measure was developed using information revealed after the matching between candidates and positions occurred and therefore was not observable by candidates before they submitted their rank-ordered list of job postings. The data used to construct the two measures were released by the Brazilian government in text format and were publicly available. To separately identify the

impact these two measures of competitiveness in candidates' revealed preferences, I created a novel dataset by web scrapping the administrative data on the applicants' demographics, exam scores, and list of preferences for the jobs, and also the content of the job description for each job posting in their original text format, and subsequently merged all this information into a single database containing variables linked to job postings' attributes, candidates' characteristics, and interaction between features from both candidates and job postings.

Using this novel dataset, I estimate how much high-performing candidates are willing to pay, on average, to expand the number of openings available in each job posting and to increase their matching probability. To do that, I estimate a model to recover the linear coefficients that capture exam takers' preferences for the two measures of competitiveness and other job attributes. With these coefficients, I compute the applicants' willingness to pay (WTP) for the two measures of competitiveness and other job attributes offered by each job posting, which allows me to discover the relative importance that candidates give to money in comparison to job attributes of interest.

This study obtains several substantial findings. The number of openings linked to the job postings and the mean value of candidates' matching probabilities for each particular number of openings are positively correlated. This pattern is consistent with the conjecture that an increase in either variable indicates that the job posting is less competitive. When choosing their most preferred alternative, high-performing candidates are willing to pay, on average, 3.6 percent of the earnings to increase their matching probability by 10 percentage points and 2.3 percent to have 10 additional openings in a job posting. In situations where they are ordering a set of alternatives, these candidates would give up, on average 5.7 percent of the earnings to increase their matching probability by 10 percentage points and 0.2 percent to have 10 additional openings in a job posting. The findings are consistent with a competition avoidance mechanism, and after controlling for the type of occupation, there is no evidence that the candidates are willing to trade off money for prestigious, hard-to-get jobs. Furthermore, the results are robust to excluding the bottom part of the rank-ordered lists, which could be noisier, from the analysis and to introducing heterogeneity in the candidates' demographic types.

Specifically, I develop a model of the highest matching probability that a candidate could obtain for a particular job posting, for a given cutoff score, their academic and professional titles, and a distribution of their components' scores in the exam, which I name max-matching probability. I then estimate the parameters of the components' scores distribution using a constrained Maximum Likelihood Estimator and apply a plug-in estimator to obtain the estimated probabilities. To address endogeneity concerns, I leverage quasi-experimental variation from individuals' exam scores and construct a leave-one-out instrument for the max-matching probability associated with each pair of candidate and job posting. As a preliminary empirical analysis of whether either the max-matching probability or the number of openings could influence the ranking decisions of the candidates, I estimate a control function logit model using the entire sample of matched and waitlisted candidates, where the dependent variable indicates whether the job posting is the candidate's most preferred option or not. To determine which attributes could influence candidates' decisions more broadly,

in addition to the choice of the best alternative, I build on the standard exploded logit model (Allison and Christakis (1994)) and incorporate the revealed preferences from observed rankings of less preferred alternatives into my analysis while also handling censoring and selection issues in the dataset. Lastly, using the parameters estimated from both models, I adopt the willingness to pay measure derived in Maestas et al. (2023) to compute what fraction of the earnings advertised in the job postings the candidates are willing to pay, on average, to have or increase an attribute.

The remainder of this paper is organized as follows. Section 2 presents the literature review and contributions to the literature. Section 3 introduces the institutional background for the centralized civil service exams in Brazil. Section 4 describes the raw data sources and the methodology used to convert them into a new database. Section 5 develops a model of the highest matching probability for each pair of candidate and job posting. Section 6 describes the estimation procedures implemented to estimate the willingness to pay coefficients. Section 7 reports the WTP estimates for the two measures of competitiveness and other job attributes of interest. Section 8 investigates how certain aspects of the data could influence the WTP estimates. Section 9 concludes.

2 Literature Review and Contributions

This paper is closely related to two strands of the literature. First, it contributes to the growing literature on revealed preferences approaches applied in the labor market setting. In this study, I use a revealed preferences approach to estimate compensating wage differentials. A more traditional approach is to use hedonic wage regressions, which often relies on the assumption of competitive equilibrium in the labor market (Rosen (1974), Rosen (1986)). In the context of public sector jobs, the government has a lot of control in setting the wages, leaving little to no room for bargaining over wages and installing an imperfect competition setup. Another well documented obstacle to identification are unobserved factors that can affect wages[1]. A way to circumvent these problems and generate reliable estimates is to implement novel approaches.

A new wave of studies tries to overcome the identification issues - such as imperfect competition settings and the existence of unobserved factors that can affect wages - of estimating compensating differentials applying revealed preferences methods. Most of them use experimental data[2], since they can obtain random or quasi-random variation in levels of job amenities. However, a challenge for most experiments is how to deal with inattentive participants, even in the presence of several attention checks (Krefeld-Schwalb et al. (2023)). An alternative is to emulate a real-life job search process, as in Mas and Pallais (2017). In their paper, the experimenters applied to job openings through a well-known job search platform. Another is to restrict the analysis to a particular group of people who are known to answer surveys in a more attentive way, such as undergraduate students from a selective university, which was the approach taken by Wiswall and Zafar (2018). An intrinsic benefit of the CPNU data in comparison to the majority of the experimental data is that the

[1] See Lavetti (2023) and Manski (1987) for a deeper discussion about how unobserved factors can affect the estimation of compensating differentials.

[2] An exception is Sorkin (2018), which uses a revealed preferences approach with administrative data.

candidates who took the civil service exams, particularly those who are high-performing and ended up being either matched or waitlisted, were exposed to a real-world high-stakes environment and revealed their preferences for job postings through a ranked-ordered list. Moreover, I build on Chapman and Staelin (1982) and implement a model that handles the trade-off between attributes along the entire rank-ordered list, preserving the data generating process of the data while accounting for its censored[3] nature, and accessing which job attributes could influence candidates' decisions in a more general sense, not only their first choices. Finally, the novel rank-ordered approach combined with the willingness-to-pay measures proposed by Maestas et al. (2023) provide reliable estimates even in the absence of a large sample.

The second body of work to which this paper is related focuses on the effects of (perceived) competition on individuals' decisions. There is a large literature in behavioral and experimental economics documenting the effects of competitiveness in a more general sense, particularly their heterogeneous effects between genders (Niederle and Vesterlund (2007)). Recently, a few papers analyzed the effects of competitiveness on labor market outcomes. Buser et al. (2014) links responses to competitiveness with real-world educational decisions, which have consequential implications for labor market outcomes. They find that among students with no observable differences in ability, gender differences in competitiveness account for a sizable portion of the gender gap in choosing more prestigious academic tracks. In the CPNU job posting choice setting, in contrast, individuals' demographic heterogeneity does not seem to play a relevant role among the individuals with similar observable measures of ability (high exam scores), although I cannot observe their gender. Banfi and Villena-Roldan (2019) provides evidence that high-wage job advertisements on online job boards attract more applicants. However, Belot et al. (2022) ran a field experiment in which they document that a sizable fraction of job applicants revealed interest solely in low-wage job postings. To understand the main driver of this sorting pattern, they conducted a survey with external raters, and their perception was that high-wage jobs were more competitive. Buser et al. (2026) explores two measures of willingness to compete, one elicited through a laboratory experiment and the other via an unincentivized survey question. The study finds that willingness to compete strongly predicts monthly gross income and occupation, conditional on education level. Relatedly, I use monthly earnings and type of occupation as controls to analyze whether my two measures of competitiveness in getting a job affect candidates' preferences for positions in a real-world, high-stakes setting, where all individuals in my sample have college degrees.

Other measures of perceived competition are also studied in the context of job applications. An example is the number of applicants that a job posting receives, a piece of information that is becoming largely available with the advent of online job platforms, such as LinkedIn. Gee (2019) implemented a field experiment on this platform that randomly assigned users actively looking for a job in two groups: one could see the number of applicants available in each job posting and the other could not. They find evidence that a job seeker is more likely to finish the application if they see the number of applicants in the job post. Fradkin et al. (2025) go one step further and investigate

[3]The candidates' ranked-ordered lists of job postings are only partially observed. For details see Section 4.

the mechanism through which information on the number of applicants can affect the job search. On the one hand, a job posting with many applicants could signal a high degree of competition, but its popularity could also indicate that the job is high quality. They concluded that applicants prefer to avoid competition by refraining from applying to postings with many prior applications and redirecting their search to postings with few applications.

In contrast, in the CPNU setting, before applying to the job postings, job candidates have access to information about how many openings each posting offers rather than the number of applicants. Altmann et al. (2026) presents a closely related environment, an online job search platform hosted by the Danish government. In one arm of their field experiment, the treated group of job seekers is exposed to the number of job openings in their selected set of occupations, which are disclosed by the candidate when registering on the platform, while the control group is not. The treated candidates increased the fraction of applications sent to postings related to the occupations they selected compared to the control group. In the Brazilian platform analyzed in this paper, every candidate has access to the number of openings of the job postings associated with each occupation. Additionally, I can observe not only the jobs the job seeker applied for but also the order of preference. Hence, I examine the impact of the number of openings in how the candidates rank the postings they apply for, instead of the decision of applying or not.

Specifically, this paper examines the individual effects of two job attributes, the number of openings, and the max-matching probability, when they occur simultaneously. In the seminal work of Mortensen and Pissarides (1994), the probability that an unemployed worker finds a job is given by the ratio between the number of worker-firm matches and the number of unemployed workers. Since every opening was matched to a candidate in the CPNU context, one could apply this concept and analyze the ratio between the openings of a job posting and the sum of all candidates who applied for it. However, this approach would not be suitable in a situation where candidates apply for multiple jobs at the same time through an Deferred Acceptance algorithm. As a simple illustration, suppose that there are two jobs postings, A and B, each one with the same number of openings. If every candidate includes both job postings in their rank-ordered lists, the ratio defined above would be the same for A and B. Nevertheless, if job A was ranked first by all candidates, it would be harder for a candidate to be matched with an opening associated with it. In other words, the candidates matched with job posting A would exhibit higher exams scores.

To address this issue, I developed a model for the probability that a candidate would be matched with a particular job posting if they had ranked it as their first option, given their score. I named this probability the max-matching probability. This probability takes into account the difference between the candidates' scores and the cutoff score (minimum score of a matched candidate) for each job posting. Since the cutoff scores depend on the candidates' submitted preferences, I borrowed the leave-one-out cutoff score methodology from the school choice literature (Figlio et al. (2023), Idoux and Corradini (2024)) and constructed an instrument for the cutoff scores leveraging quasi-experimental variation from candidates' observed scores. After accounting for the endogeneity issue, I estimate candidates' willingness to pay to have a higher max-matching probability. The estimates

show that candidates are, on average, willing to give up a fraction of the earnings from the job postings to have a higher max-matching probability.

The estimates' results on the number of openings move in the same direction. On average, candidates are willing to forgo a fraction of the earnings to have more openings in the job postings they applied for. One possible mechanism that could be driving this behavior is competition avoidance, which would be in line with the findings in Fradkin et al. (2025). Preference for prestigious, hard-to-get jobs (Zhan (2015)) is less likely to serve as a key determinant of the results, particularly after controlling for the type of occupation. Other studies, such as Dreyfuss et al. (2026), find that in candidate proposing DA settings, where preferences include expectations-based reference dependence, the competition avoidance behavior is also observed.[4]

3 Institutional Background

Civil service exams have been widely implemented in the selection of public servants in Brazil, especially in recent decades, and currently account for about 80% of admissions to the public sector in the country (Mocanu (2024)). However, the admission of these government officials was carried out in a decentralized manner, with each public agency responsible for implementing its own admission exam. The *Concurso Público Nacional Unificado* (CPNU) first edition took place on August 18, 2024. The main purpose of creating a unified national exam is to have a unique selection process, occurring simultaneously in dozens of cities across the entire country, which selects applicants for various job openings. The exam selected 6,640 employees for federal public jobs in 21 different public bodies[5], from 970,037 applicants who took the exam (2,144,435 individuals signed up). Instead of taking an exam for one particular job posting, each candidate chose one of the eight theme blocks: (1) Infrastructure, mathematics, and engineering; (2) Technology, data and information; (3) Environmental, agrarian and biological; (4) Server work and health; (5) Education, Health, Social Development and Human Rights; (6) Economic sectors and regulation; (7) Government management and public administration; (8) High School level. Each theme block offers a set of job postings, and within the chosen block, candidates can rank all the job postings they want. After every candidate has submitted their rank-ordered lists (ROL) through an online platform, they take the exam for the specific block they have chosen. Once every exam is graded, each job posting provides a final score based on the exam's score[6] for each candidate who ranked it in their lists, and the candidates are assigned to the openings through candidate-proposing Deferred Acceptance Algorithms, which run independently within each theme block.

[4]In Section 3, I discuss why sub-optimally trying to deviate from truth-telling to "game the system" is an implausible explanation for the behavior of the candidates analyzed in this paper, although it is a valid concern in other contexts (Rees-Jones (2018)).

[5]19 of these public bodies appear in the *Anexo I* of at least one Theme Block's *Edital*. Two public bodies, *Ministério da Educação (MEC)* and *Ministério dos Direitos Humanos (MDH),* appear only in the description of one job position in Theme Block 5 *Edital* listed as being corresponding to MGI in *Anexo I*.

[6]See Section 4 for a complete discussion on the score of the exam by its components and the final score for each job posting.

The Deferred Acceptance Algorithm implemented in every theme block generates the final match between candidates and job openings, and it depends on both the candidates' final score for each job posting and their submitted rank-ordered list of job postings. The candidates are assigned to an opening of their most preferred *feasible* job posting, i.e., the most preferred job posting among the ones in which the candidate has a final score greater or equal to its respective the cutoff score[7]. An important caveat is that in the CPNU setting there are affirmative action quotas, resulting on a mechanism slightly different from the standard DA without reserved openings (Aygün and Bó (2021) and Bó et al. (2024) explain the modifications relevant to this setting in comparison to the standard version of the well-known incentive compatible candidate-proposing DA, thoroughly described in Roth (2008)).

The cutoff score for each job posting is also based on its number of openings. The number of openings is not only defined before candidates submit their preferences, but is also determined by the budgetary and financial availability of the federal government and the demands of each particular public body (Brasil. Ministério da Economia (2019)). Therefore, the number of applicants and their preferences do not affect the number of openings for each job posting in this setting. Additionally, the high volume of candidates for each job posting ensures that all positions are filled, and the ones designated for affirmative action are filled exclusively by candidates who meet its respective eligibility criteria. According to the exams' rules, the candidates whose scores for a particular job posting are high enough so they are matched to a general competition openings while also being eligible for openings designated to affirmative action will not count for the purpose of filling reserved openings, implying that the cutoff scores for openings reserved for affirmative action are always lower or equal than the cutoff scores for the openings reserved for general competition.

A document named *Edital* carefully describes the exam, the hiring process, and the job postings themselves. The entire content of the *Edital* will be elucidated in section 4. Nevertheless, it is worth mentioning what is made salient in those *editais* and what is left out. Many attributes of each job posting such as the public body that it is associated with, the monthly earnings, location, type of occupancy, number of openings for each job posting and the number of expected hours of work per week are highlighted in the documents and often appear more than once. Other important attributes frequently mentioned in the literature on non-wage amenities[8] such as retirement contributions, health insurance, and paid time off do not appear in the job descriptions offered in CPNU because these benefits are defined by *Law 8,112* and have a common basis for all federal civil servants hired through civil service exams. Additionally, there are various ways to categorize public bodies within the Brazilian federal administration. According to the current Federal Constitution, one of them is the distinction between direct and indirect administration. The public bodies classified as the former, such as the Ministries, are directly linked to the Union, and the ones classified as the latter are only indirectly linked to the Union and have more autonomy to execute public policies, even

[7]In some cases, there are ties between candidates' final scores for the same job posting. There are specific rules on how the allocation will proceed in the advent of a tie. However, it is important to note that the tie-break rules are exogenous to the candidates preferences.

[8]See Sockin (2022) and Maestas et al. (2023) for a comprehensive catalog of non-wage amenities.

though their budgets are a part of the main budget of a public body in the direct administration.

Another instance of the *edital* and the online platform provided by the for the candidates to submit their rank-ordered lists is that they were designed to make people disclose their true preference order. In particular, there are many reasons why it is reasonable to assume that the candidates are telling the truth. The first versions of the *editais*[9] for every Theme Block were released on January 10, 2024 and contained the deadline for the submission of the order of preferences for the job postings, on February 09, 2024. Hence, every candidate had one month to make their ranking decisions, which is a long time, especially compared to experimental settings. Moreover, not only is the time to make the ranking decision long, but the number of alternatives is also low. There is vast documentation (Chernev et al. (2015)) that facing a high number of options could induce cognitive overload in the candidates, jeopardizing their ability to rank all options truthfully. In the CPNU context, each job posting has a specific college major or academic specialization requirement, which reduces the size of candidates' feasible choice set and mitigates their cognitive overload. In practical terms, a candidate who chose theme block 7, which has the highest number of job postings, would have to have at least 2 (3) different major degrees to be eligible for more than 20 (30) out of a total of 43 vacancies offered by different job postings.

The government also presented a set of instructions in the *edital* that unravel the best practices for the submission of preferences. Over the course of many subsection items, the document reiterates that submitting the true preference order is the optimal ranking strategy for the candidate. In addition, there was also an attention check on the actual online platform where the candidates submit their rank-ordered lists. After ranking every option, the candidates must check a box to acknowledge that they are aware that upon being approved for a particular position, based on their score, they will no longer be eligible for other, less desirable positions[10]. Lastly, the candidates, specially the high-performing ones, were exposed to an extremely high-stakes environment. The rate of applicants per opening was higher than 100 for every theme block and could be as high as an average of more than 1,000 applicants per vacancy. Furthermore, the exam itself was costly for the candidates, who had to pay the exam's fees or request a payment exemption, commute to an accredited exam site, and solve the exam's questions for many hours. Besides, those who were matched or waitlisted are likely to have spent a lot of time studying and preparing for the exam. Therefore, the participants in our sample have a strong incentive to submit a ROL that truthfully represents their preferences.

In terms of advantages, the Brazilian Ministry of Management and Innovation in Public Services (MGI) states that the three main benefits of creating a unified national exam for the government are: (i) *Economies of scale*: The cost per unit for the exams of each type of vacancy is decreased from the examiner's perspective since for many different types of vacancy, the same exam will be offered. (ii) *More effectiveness*: Since the candidates can access a broader set of vacancies, they can choose the ones that fit their preferences better, avoiding the rent-seeking behavior of taking any

[9] All versions of the *editais* are made publicly available by the Brazilian government.

[10] Figure 3 provides a screenshot of the attention check.

civil service exam that is available to them only to secure a high paid stable job. (iii) *More equality*: The exam was administered simultaneously in 228 cities across Brazil and 94% of the country's population were within 100 km of an exam site. Hence, virtually every citizen in the country would be allowed to take the exam and compete for a particular type of vacancy if they wanted to.

Finally, the three major advantages of creating a unified national exam for the candidates reported by the MGI are: (i) *Cost efficiency*: Each candidate only needs to pay for one application and can compete for multiple vacancies. (ii) *Vocational choice*: Allows candidates to focus on the job postings that better align with their preferences. (iii) *Transparency in the Affirmative Actions*: Have clear rules for affirmative action, which is helpful for the candidates who are eligible for them.

4 Data

4.1 Raw data sources

The MGI publicly released two types of data in text format: documents with the entire description of each job posting and individual-level data. For each of the eight theme blocks, there are PDF documents, the *edital*, carefully describing the exam, the hiring process, and the job postings themselves. These documents have around 70 pages each. Regarding the information about the vacancies, each of job posting includes the monthly remuneration, location, specialization and/or major required, number of hours per week, the public body that is offering the vacancy, the number of openings reserved for each segment of the population (general competition, black, person with disabilities, and indigenous) and activity description. Throughout the paper, there will be a distinction between the job posting and the job openings. Each job posting has a particular description in one of the eight *edital*, and multiple job openings can be offered with the same description. In this study, I will focus solely on theme block 7, which is the theme block with the highest number of job openings and job postings (see Tables 7 and 8) and offers positions related to government management and public administration. The main reason to focus on this theme block is to ensure there is enough variation in the job attributes in the job postings present in the candidates' choice set.

Additionally, the *edital* characterizes the weighting scheme used by the public bodies to assign candidates to each job opening. In general, the weighting scheme serves the same propose as a priority index. It is an ex-ante concept used in two-sided matching, where one side informs the other side about the procedure they will use to rank the agents on the other side. In our case, one side are the public bodies offering openings for the Centralized Civil Service Exams, and the other side are the job candidates. The exam has three components: (i) discursive, (ii) specific knowledge, and (iii) general knowledge. The final score that each candidate has for each job posting that they included in their ranked ordered lists is a weighted average of the score in the three components and a job posting specific score for titles. Each job posting defines their weighting scheme and how many points each title is worth, and this information is available in the *edital* to candidates before they submit their preferences. The way in which public bodies outline the weighting scheme

is not the same for each job posting, but there are only up to three ways to rank within theme block 7. Each way assigns different weights to the scores of each part of the exam and to the titles (academic and non-academic), as described by Table 9. Based on the weighting scheme (priority index), the candidate can be either matched, waitlisted or rejected. A rejected candidate cannot be waitlisted and a matched candidate can only be waitlisted for jobs with a better rank in their submitted rank-ordered list than the one they were matched with.

The second type of data is the individual-level data. Candidates must submit their personal information, which appears in the *resumo da inscrição* (registration summary), before taking the exam. The *resumo da inscrição* for each candidate contain personal information and demographics such as the individual's name, a Personal Identifier Number (PIN), their date of birth, their gender identity, their marital status, children's age bin (if they have children), their educational level, where did they take the exam, both state and city, the theme block they chose, the rank-ordered list of the job postings within the theme block they have chosen, and their eligibility to at least one of the following types of affirmative actions: black person, a person with disabilities, and Indigenous person. Even the candidates that were not matched with any opening have submitted their *resumo da inscrição*.

However, the complete information on the registration summary is not publicly available, and the Brazilian government does not provide access to it. What is publicly available in terms of individual-level data is, for each job posting in each theme block, a PDF file[11] with a few characteristics of the exam takers who were either accepted or waitlisted for an openings from a particular job posting. These characteristics include the exam takers's PIN, their birth date, three dummy variables indicating if they classify for any of the affirmative action categories (black, person with disabilities, indigenous), a dummy variable indicating if they were accepted or waitlisted, their scores for each part of the exam, and the numerical rank of the job posting in terms of the exam taker submitted preferences. Hence, I cannot observe the full rank-ordered lists submitted by the candidates, only observe their partial lists with the jobs that they were either accepted or waitlisted for.

4.2 Dataset construction

4.2.1 Job postings' characteristics

Monetary incentives are one of the most important characteristics of a job offer. However, a large part of the decision of which job to accept or which job is preferred to the other consists of evaluating the non-wage amenities attached to them (Mas (2025)). I extracted the observable job characteristics from each job posting in the *edital* and matched them to their respective job posting ID. As stated previously, each job posting includes the monthly remuneration, location, specialization and/or major required, number of hours per week, the public body that is offering the openings, and activity description. In addition, the documents present how many openings are offered in each job

[11] The files can be accessed here: `https://www.gov.br/gestao/pt-br/concursonacional/resultado/resultados`.

posting, i.e., how many job with the same characteristics are available, and how many openings for each job posting are reserved for general competition, black individuals, person with disabilities, and indigenous people. Table 10 provides a summary of these characteristics. In the reminder of the subsection, I explain how each one of these characteristics is analyzed and transformed into variables used in the estimation procedures.

Monthly remuneration. All compensations appear in the description of the *edital*. All values are on a monthly basis and in Brazilian real - the official currency of Brazil - of 2024. The types of compensation described are the base pay, the specific bonus, the performance bonus, and the remuneration for qualification. The base pay appears in every job posting description and each one of the other types of compensation may or may not appear in a particular description of a job posting. The specific bonus is a compensation related to either the task that the job posting demands or its correspondent public body. The performance bonus is a compensation based on a grade scale from 0 to 100. All performance bonuses in our sample are normalized to its value of 80 points, which is the value that the applicants matched to a vacancy that offers a performance bonus will receive before their first performance evaluation. Lastly, each job posting in each theme block presents at most three types of remuneration for qualification. For theme block 7, which require a college degree, there is a remuneration for non-master's specialization, for master's degree and for doctoral degree, and the higher is the final degree, the higher is the remuneration. For each job posting, I define as its **total earnings with no titles** the sum of the base pay, the specific bonus, and the performance bonus; and the total earnings with titles is the total income with no titles plus the highest possible value of the remuneration for qualification, which corresponds to the doctoral degree.

Hours per week. The vast majority of job postings has a 40 hours per week requirement. Only one posting in the sample have a different requirement, of 25 hours.

Educational requirement. One factor that plays an important role in how an individual values a certain job is their skill level, both in broad terms such as if the applicant is a college graduate or not, and in specific terms such as their college major if they have one. The documents with the job descriptions are informative in both aspects. First, all theme blocks, except for Theme Block 8 - which requires only a high school degree - require the candidate to have a college degree to be eligible for any type of vacancy within the theme block[12]. Secondly, each theme block from 1 to 7 is associated with a main area of knowledge, so that the vast majority of job postings within each of these theme blocks require a college major correspondent to a specific area of knowledge[13]. Theme block 7, which I will focus on, require specific majors related to government management and public administration for 30 out of 43 job postings (almost 70 percent). For the other 13 job postings, the candidate could have any college degree.

[12]A candidate who holds a college degree is allowed to choose Theme block 8.

[13]To be specific, all theme blocks from 1 to 7 have at least one type of vacancy for which the requirement is "Any area of knowledge", meaning that a candidate with any college major is eligible for that particular type of vacancy provided that they have a high enough priority index. The other types of vacancy in the same theme block require a college major that is associated with the area of knowledge of their theme.

Activity description. Another way to measure the skill level of an individual is to analyze the main tasks required in their job. In our dataset, the main tasks required for each job vary depending on the occupational category. The International Standard Classification of Occupations (ISCO) defines occupation as a *"set of jobs whose main tasks and duties are characterized by a high degree of similarity"*. To adhere to this definition and to make the **occupation categories** comparable between job postings, I created a new set with broad occupational categories to classify all job positions available in the CPNU[14]. There are 15 different job positions listed in theme block 7's *edital*. Some of these positions appear more than once, sometimes requiring different specialties/majors or being offered by different public bodies. In the paper, each of these job positions is classified into one of five broad occupational categories. Each broad occupational category is a set that contains at least one job position, and the occupational categories are disjoint sets that form a partition of all the job positions. To construct these sets, we extracted the main role of each job position, which is displayed as the first component of the job position's title. The broad occupational categories are analyst, specialist, researcher, technician, and same as major. To be classified as the same as major, the job position has to be a direct reference to the specialty/major, e.g. Accounting or Business.

Location. In the *edital* description of each job posting, there is the location where the activities will be executed. In this paper, the term location refers to a set that contains at least one city, but could contain more than one city. The CPNU setting presents a distinctive feature: a large fraction of the jobs offered possesses a location that encompasses many cities in their description, which means that the job applicant who includes a job posting with many cities in their ROL does not know for sure in which city they would be appointed to in case they were matched to one of its openings. There are only two cities that appear in unitary location sets: *Rio de Janeiro* and *Brasilia*, the latest being the capital of Brazil. The other sets, which contain more than one city, are defined as *Many cities*[15].

Public body. Every job posting in the CPNU is offered by a specific public body in the federal government, which is explicitly stated in the job description. There are 14 different public bodies in the dataset, 8 of them have only one job posting, and of the other 6, and even though they have multiple job postings, only one public body offers a different remuneration for different job postings (see Table 11). Additionally, in this paper, public bodies are classified into two sets: those that are **Ministries** and those that are not. Roughly 40 percent of the openings are from Ministries, and the vast majority of the job postings offered by them are located in the capital of the country.

Number of openings. Each job posting has the number of openings, i.e., the number of vacancies available with the exact same description, stated in the *edital*. The second column of Table 12 displays the total number of openings available for each job posting. They range from 1 to 238 and even within the same public body or the same type of occupation, there is a lot of variation in the job postings' number of openings. The share of openings reserved to candidates who are eligible for affirmative action in each job posting, which are defined by federal laws and pre-determined at

[14] Appendix D shows the construction of each set of broad occupational categories, with their corresponding job positions.

[15] For a detailed discussion about all the locations sets and its elements, see Appendix C.

the job posting description along with the total number of openings. Almost three-quarters of all openings are reserved for general competition, which means that every candidate is eligible for those openings. Every job posting with 4 or more total openings has to reserve a share of its openings for black candidates, every job posting with 5 or more total openings has to reserve a share of its openings for candidates with disabilities, and only one public body (National Foundation for Indigenous People - FUNAI) has openings reserved for indigenous candidates, and they represent less than one percent of the total openings for all postings. Overall, almost twenty percent of the openings are reserved for black candidates and six percent of the openings are reserved for people with disabilities.

Exam weighting. Each job posting provides one of the three weighting schemes (Table 9) for the candidates' scores in each component of the exam, and I use them to estimate the distribution of the final score for each candidate in each job posting. In addition, to compute the distribution of the final score, I also use the candidates' scores in each component of the exam, which I discuss in more detail in the following section.

4.2.2 Job applicants' characteristics

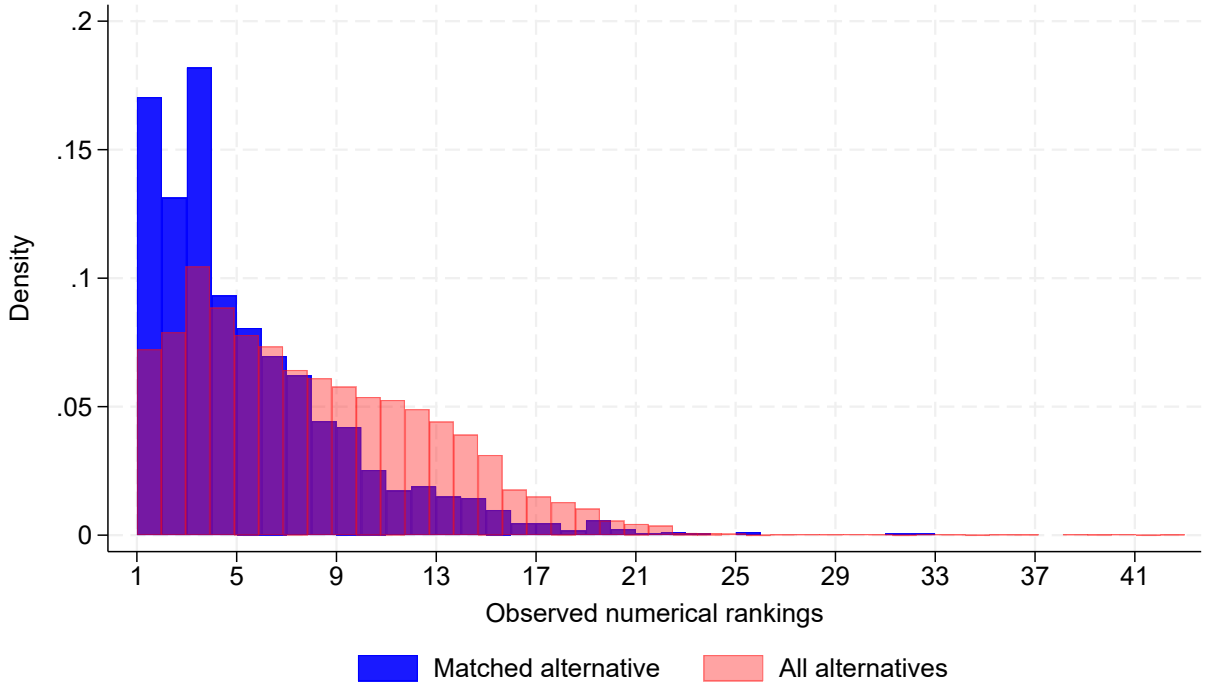
Preferences. After the deferred acceptance algorithm assigns the candidates to the jobs based on their rank-ordered lists and priority index, the government released a file for each job posting, which contained information of all candidates who were either matched or waitlisted for one of the job openings offered by this particular job posting. The same candidate had the same PIN in all files, which enabled me to partially recover the rank-ordered list of each candidate, since I can observe the numerical rank of the job posting in terms of the exam taker submitted preferences. Hence, I construct a new rank-ordered list for each individual that includes only the jobs that the candidate was either matched or waitlisted, preserving the order of the observed ranked alternatives.

When the candidates are matched to a job posting, I can only observe the options they ranked as preferred to the one which they were matched to[16]. Consequently, when the candidate is matched to their first option, I can only observe their most preferred alternative. Figure 1 displays the histograms of the observed numerical rankings for all observed alternatives and for the ones that were matched to a candidate. The majority of the matched candidates were matched to an alternative within their top five options and less than ten percent of all observed numerical rankings, including those for which the candidates were waitlisted, are above the 15th position in their rank-ordered lists. There are 1,737 matched candidates, which is equal to the total number of openings offered.

As mentioned above, there are two types of censoring in the data: (1) I can only observe the numerical rankings of alternatives that the candidates were either matched or waitlisted for, and (2) for the matched candidates, I cannot observe any alternative that was ranked as less preferred to the one that they were matched to. Table 1 presents the summary statistics for the number

[16]This feature of the data is a product of the matching algorithm, which only allows matched candidates to be waitlisted for jobs that they ranked as preferred to the one to which they were matched.

Figure 1: Distribution of rankings



Notes: The figure displays histograms of the observed numerical rankings. In red, it includes all observed alternatives for the matched and waitlisted candidates. In blue, it only includes the numerical ranking for matched candidates, and only the alternative that they were matched to.

of candidates, according to their self-declared demographic characteristics.[17] With the exception of indigenous candidates, who represent only 0.7 percent of the total candidates, the sample with candidates from whom I can observe two or more ranked alternatives and are unmatched possess a demographic distribution analogous to the sample with all candidates. Henceforth, I will refer to the sample with all but indigenous candidates ($N = 5,174$) as the full sample. The number of observed ranked alternatives is also balanced across demographics characteristics (see Table 13).

Components' scores. I can observe the weighted score for each component of the exam of every candidate that was either matched or waitlisted in a particular job posting. Using the weighting scheme implemented by the job posting, I normalize the components' scores of all candidates in the sample from 0 to 100 and obtain the raw exam components' scores. Note that regarding the exam, each candidate has only three different scores, one for each component. Figure 4 shows the distribution of the normalized scores of the candidates in the full sample.

Titles. On the other hand, for the titles, each job posting will give each individual a distinct score based on the candidates' qualifications related to the needs of that particular posting.

[17]Note that some individuals self-declared simultaneously as black and person with disability.

Table 1: Number of candidates - summary statistics

	Only one observation or matched	Two or more observations & unmatched	Total
All candidates	2,637 (50.6%)	2,574 (49.4%)	5,211 (100.0%)
No affirmative action			
No	769 (29.2%)	698 (27.1%)	1,467 (28.2%)
Yes	1,868 (70.8%)	1,876 (72.9%)	3,744 (71.8%)
Black			
No	2,042 (77.4%)	2,023 (78.6%)	4,065 (78.0%)
Yes	595 (22.6%)	551 (21.4%)	1,146 (22.0%)
Person with disability			
No	2,477 (93.9%)	2,408 (93.6%)	4,885 (93.7%)
Yes	160 (6.1%)	166 (6.4%)	326 (6.3%)
Indigenous			
No	2,600 (98.6%)	2,574 (100.0%)	5,174 (99.3%)
Yes	37 (1.4%)	0 (0.0%)	37 (0.7%)

Notes: The table presents the summary statistics for the number of candidates. For each sample selection, the *Yes* and *No* percentages are with respect to each demographic group.

4.2.3 Cutoff scores

In a Deferred Acceptance algorithm, whether a candidate gets matched to a particular job posting or not depends on their own submitted rank-ordered list, the other candidates' rank-ordered lists, the scores for that job of every candidate that included it in their rank-ordered list, and the number of openings available in that particular job posting. The realized cutoff score for each job posting is computed as the minimum final score attained by a candidate matched to it. Specifically, for each job posting, the minimum score associated with candidates matched to each type of opening considered in the sample (general competition, black, and person with disabilities) is used to compute its cutoff score. As a result of the matching mechanism, the cutoff score for general competition for a particular job posting is always greater than or equal to its cutoff scores for openings reserved for affirmative action. On the other hand, there are no rules determining whether the black cutoff score or the PwD cutoff score should be higher for a job posting that has openings reserved for both, even though the cutoff scores for black candidates are predominantly higher in the data.

5 Max-matching Probability Model

Let $\mathcal{I} := \{1, \cdots, I\}$ be the set of individuals applying for jobs and $\mathcal{J} := \{1, \cdots, J\}$ be the set of all job postings available. $Y_{ij} \in \{1, \cdots, J\}$ is the numerical rank given to job j by applicant i. Preferences are stated in descending order, i.e., 1 is the value assigned to the best choice, 2 is the value assigned the second-best choice, and so on until J, which is the value assigned to the worst choice. Let K be the number of components of the exam. For each component k, candidate i receives a score s_i^k. Assume that the underlying process of each component's score can be approximated by

a random-effects latent ability model:

$$s_i^k = \delta^k + \lambda_k a_i + \varepsilon_i^k \tag{1}$$

where δ^k is the expected value of component k's score, a_i is the candidate's latent ability, and ε_i^k is an error-term. Assume that $a_i \sim \mathcal{N}(0, \gamma^2)$, $\lambda_k \in \mathbb{R}$ is a factor loading, $\varepsilon_i^k \sim \mathcal{N}(0, \sigma_k^2)$, and that the latent ability is independent of the error-term and the error-terms are independent across components. Let $\boldsymbol{s}_i = (s_{i1}, \ldots, s_{iK})^\top$ and $\boldsymbol{\delta} = (\delta_1, \ldots, \delta_K)^\top$. The normality and independence assumptions imply that, for each individual i, $\boldsymbol{s}_i \sim \mathcal{N}(\boldsymbol{\delta}, \boldsymbol{\Sigma})$, where the covariance matrix $\boldsymbol{\Sigma} \in \mathbb{R}^{K \times K}$ has diagonal entries $\Sigma_{kk} = \lambda_k^2 \gamma^2 + \sigma_k^2$, off-diagonal entries $\Sigma_{kk'} = \lambda_k \lambda_{k'} \gamma^2$, and can be written as $\boldsymbol{\Sigma} = \gamma^2 \boldsymbol{\lambda}\boldsymbol{\lambda}^\top + \boldsymbol{D}$, where $\boldsymbol{\lambda} = (\lambda_1, \ldots, \lambda_K)^\top \in \mathbb{R}^K$ and $\boldsymbol{D} = \operatorname{diag}(\sigma_1^2, \ldots, \sigma_K^2)$.

In each job posting j, each candidate i receives a score $t_{ij} \in \mathbb{R}$ for their academic and professional titles, which is known by the candidate before they submit their preferences. The final score s_{ij} that candidate i receives in each job posting j is a weighted average across all K components' scores and the title score:

$$s_{ij} = \sum_{k=1}^{K} \pi_{k,j} s_i^k + \left(1 - \sum_{k=1}^{K} \pi_{k,j}\right) t_{ij} \sim \mathcal{N}\left(\boldsymbol{\pi}_j^\top \boldsymbol{\delta} + (1 - \mathbf{1}^\top \boldsymbol{\pi}_j) t_{ij}, \boldsymbol{\pi}_j^\top \boldsymbol{\Sigma} \boldsymbol{\pi}_j\right) \tag{2}$$

where $\boldsymbol{\pi}_j = (\pi_{1,j}, \cdots, \pi_{K,j}) \in \mathbb{R}_+^K$, $0 < \sum_{k=1}^K \pi_{k,j} \leq 1$ and every candidate knows the weighting scheme of each job posting j before submitting their preferences.[18]

Define the cutoff score for job posting j, c_j, as the minimum final score attained by the candidates matched to j. Let $q_j \in \mathbb{Z}^+$ be the number of openings available in job posting j. Assume that each candidate knows the rules of the assignment mechanism, the number of openings for each job posting, the distribution of their exam scores for each job posting, and the distribution of all candidates' submitted rank-ordered lists and exam scores. Assume that the number of candidates is large enough so that the uncertainty about the value of the cutoff scores is negligible for every job posting and take them as given.

The max-matching probability that candidate i is matched to job posting j is

$$p_{ij} = \Pr\left(s_{ij} - c_j \geq 0\right) \tag{3}$$

In light of the candidate-proposing deferred acceptance algorithm, if candidate i had ranked job posting j as their first (most preferred) option, then p_{ij} would be equal to their probability of being matched to it. The max-matching probability is the highest matching probability that candidate i can achieve for job posting j for a given distribution of their score and the cutoff score, regardless of how they ranked it on their submitted ranked-ordered list[19].

[18] In the model, there are virtually no ties in the scores of each job posting because of the continuous and random nature of the latent ability and error-term.

[19] As discussed in Kapor et al. (2020), in a candidate-proposing deferred acceptance algorithm, the rational expectation matching probability for candidate i in job posting j is $p_{ij} = \Pr\left(s_{ij} - c_j \geq 0, s_{ij'} - c_{j'} < 0 \ \forall j' \text{ s.t. } Y_{ij} > Y_{ij'}\right)$.

6 Estimation Procedure

I want to recover individual preferences from the exam takers and use the estimated preference parameters to compute their willingness to pay for the characteristics present in each job posting. As in Fack et al. (2019), I define weak truth-telling as the strategy of ranking all submitted options truthfully, i.e., according to the true preferences of the individuals, and that the number of options ranked is exogenous. However, the individual is allowed not to rank all the options available. I assume that there are no peer effects, in the sense that other candidate's matches do not affect one's utility, and that there are no rank effects, meaning that the priority index rank within the job posting does not affect the individual's utility. There are no bonuses or different pay schemes for those who rank better. The same job will be performed by all individuals matched to the same job posting. The following subsections elucidate the estimation procedure of the max-matching probability, the construction of instrumental variables for the max-matching probability and the number of openings, the reduced-form method to recover preferences applied solely to the candidates' first (best) option, the estimation of the job posting rank-ordered choice model and the derivation of the willingness to pay measure for each attribute of the job postings.

6.1 Max-matching probability

In this subsection, I present the estimation procedures for the distribution parameters of the final scores, s_{ij}, and max-matching probabilities, p_{ij}, of each candidate-job posting pair (i, j). The exam has three components: discursive, specific knowledge, and general knowledge. Hence, I impose $K = 3$. Normalize $\lambda_1 = 1$ to get identification for the final score parameters $\boldsymbol{\theta} = (\delta^1, \delta^2, \delta^3, \gamma^2, \lambda_2, \lambda_3, \sigma_1^2, \sigma_2^2, \sigma_3^2)$ and estimate them using a constrained Maximum Likelihood Estimation as discussed in Appendix E.1. Then, apply a plug-in estimator derived from the normality and independence assumptions to estimate the max-matching probabilities. From the distribution of candidate's i final score s_{ij} for each job posting j described in Equation 2, candidate's i plug-in estimator for the max-matching probability for job posting j is:

$$\hat{p}_{ij} = 1 - \Phi\left(\frac{c_j - \boldsymbol{\pi}_j^\top \hat{\boldsymbol{\delta}} - (1 - \mathbf{1}^\top \boldsymbol{\pi}_j) t_{ij}}{\sqrt{\boldsymbol{\pi}_j^\top \hat{\boldsymbol{\Sigma}} \boldsymbol{\pi}_j}}\right) \tag{4}$$

where Φ is the standard normal CDF, c_j is the realized cutoff score for the general competition openings in job posting j, $\boldsymbol{\pi}_j$ is the observed weight vector of the components' scores for job posting j, t_{ij} is observed titles' score of candidate i in job posting j. $\hat{\boldsymbol{\delta}}$ and $\hat{\boldsymbol{\Sigma}}$ are estimated with the components' scores of every candidate in the full sample through a constrained MLE.

6.2 Instrumental variable for the max-matching probability

After estimating the max-matching probability p_{ij}, the next step is to address its endogeneity issues. The endogeneity of p_{ij} is embedded in the fact that the match between a candidate and a job posting

depends on their submitted rank-ordered list, which states the candidate's preferences. Following Wooldridge (2015) and de Grange et al. (2024), I implement a control function approach to account for the endogeneity concerns.

For each job posting j, assume that the number of openings associated with it q_j, and the variables in the vector of non-monetary characteristics X_j are exogenous. Then, for each pair of candidate and job posting for which I can observe the candidate's specific score, I constructed a leave-one-out instrument similar to the ones used in recent papers in the school choice literature (Figlio et al. (2023), Idoux and Corradini (2024)). To construct the instrument, I leveraged quasi-experimental variation from individuals' exam scores and computed a new cutoff score for the job posting without taking into consideration the candidate's own realized score. Since the cutoff score of a job posting is the minimum score attained by a candidate matched to that particular job, if they were not matched, then their score is below the original cutoff and removing it will not affect the computation of the leave-one-out cutoff score. Let c_j^{IV} be the leave-one-out cutoff score. The instrument for p_{ij} is the difference between the candidate's expected score and the leave-one-out cutoff score:

$$Z_{p_{ij}} = \mathbb{E}[s_{ij}] - c_j^{IV} = \boldsymbol{\pi}_j^\top \hat{\boldsymbol{\delta}} + (1 - \mathbf{1}^\top \boldsymbol{\pi}_j) t_{ij} - c_j^{IV} \tag{5}$$

Finally, I obtain the residual for the control function approach from the following first stage regression of the matching probability p_{ij}:

$$p_{ij} = b^w \ln w_j + b^Z Z_{p_{ij}} + b^q q_j + b^X X_j + v_{ij}^p \tag{6}$$

6.3 Reduced-form first option probability

Since the max-matching probability of candidate i for job posting j is equivalent to the probability of i being matched to j if they had ranked it as their most preferred option ($Y_{ij} = 1$), a natural exercise is to analyze how the max-matching probability and the number of openings affect candidates' chances of ranking a particular job posting as their most preferred option. Formally, assume that the probability that candidate i ranks job posting j as their most preferred option, given earnings w_j, max-matching probability p_{ij}, number of openings q_j, and non-monetary characteristics X_{ij}, can be approximated by a control function logit model:

$$\Pr(Y_{ij} = 1 | \ln w_j, p_{ij}, q_j, X_j, v_{ij}^p) = \frac{e^{(\alpha + \beta^w \ln w_j + \beta^p p_{ij} + \beta^q q_j + \lambda_p \hat{v}_{ij}^p + \beta^X X_j)}}{1 + e^{(\alpha + \beta^w \ln w_j + \beta^p p_{ij} + \beta^q q_j + \lambda_p \hat{v}_{ij}^p + \beta^X X_j)}} \tag{7}$$

I estimate Equation 7 using the full sample and pooling the observed numerical rankings of all candidates. This approach allows me to preserve every observation of job postings ranked as first option in the full sample and to assess whether there is suggestive evidence of the effect of either the max-matching probability or the number of openings in candidates' ranking decisions.

6.4 Job posting rank-ordered choice model

The reduced-form presented above is an important preliminary empirical analysis of whether either the max-matching probability or the number of openings could influence candidates' ranking decisions. However, it omits information about the trade-off between attributes that can only be revealed by the distinct rankings that candidates give to their less preferred alternatives, which determines which attributes could influence candidates' decisions in a more general sense, not only their first choices, and ultimately provide reliable measures of willingness-to-pay even in the absence of a large sample (Chapman and Staelin (1982))[20].

Below, I propose a methodology that better handles the trade-off between the job postings' attributes while adapting standard estimation methods to address the challenges imposed by the data. In the dataset used in this paper, the researcher can only partially observe the rank ordered list that each job candidate submitted. To account for the censored nature of the dataset, I implement a restriction on the standard exploded logit model (Allison and Christakis (1994)). This method enables me to compare all the alternatives ranked by the individuals that are observed in a censored dataset and to preserve the information about the order, i.e., the numerical value of the rank that each job applicant assigned to each job posting. More importantly, it accommodates the partial structure of the observed ROLs and permits the comparison of observed choices made by different job applicants without having to rely on stronger assumptions.

Formally, I will follow a similar exploded logit approach as the one proposed by Allison and Christakis (1994). Assume that every candidate is playing a weak truth-telling strategy, the utility that applicant i obtains from job j is composed of a systematic component μ_{ij} and a random component ϵ_{ij}:

$$\begin{aligned} U_{ij} &= \mu_{ij} + \epsilon_{ij} \\ &= \beta^w \ln w_j + \beta^p p_{ij} + \beta^q q_j + \beta^X X_j + \epsilon_{ij} \end{aligned} \tag{8}$$

Where w_j is the monthly earnings offered by job j, p_{ij} is the max-matching probability attained by candidate i for job posting j, q_j is the number of openings associated with job posting j, $X_j \in \mathbb{R}^N$ is a vector of other non-monetary characteristics of job j, and ϵ_{ij} represents the individual unobserved heterogeneity, which I assume is i.i.d. and follows an extreme value type-I distribution. The max-matching probability p_{ij} and the number of openings q_j are as defined above. As in Allison and Christakis (1994), for each individual i, I define a random variable of the pairwise comparison between jobs j and k:

$$\delta_{ijk} = \begin{cases} 1, & \text{if } Y_{ik} \geq Y_{ij} \\ 0, & \text{otherwise} \end{cases}$$

Hence, if alternative k is less preferred than alternative j and thus receive a higher numerical rank, then $\delta_{ijk} = 1$. Assume individuals will give a better rank to the jobs that give them higher utilities.

[20] The usage of rank-ordered data poses its own challenges, which are discussed in detail in Section 8.1.

Additionally, let $P(\mathcal{J})$ be the power set of $\mathcal{J}$, and $\mathcal{J}_i \in P(\mathcal{J})$ be the observed set of alternatives ranked by individual i. Assume that within $\mathcal{J}_i$ there are no ties. Then, define the binary random variable that assumes value 1 if the choice of alternative j by individual i is observed and 0 otherwise:

$$R_{ij} = \begin{cases} 1, & \text{if } j \in \mathcal{J}_i \\ 0, & \text{otherwise} \end{cases}$$

For each applicant, their probability of submitting the censored rank-ordered list $Y_i \in \mathcal{J}_i$, as if only their observed ranked options were part of their choice set is:

$$L_i = \prod_{j=1}^{J} \left(\frac{\exp(\mu_{ij})}{\sum_{k=1}^{J} \delta_{ijk} R_{ik} \exp(\mu_{ik})} \right)^{R_{ij}} \tag{9}$$

To access the intuition for this modeling approach, assume that the individual ranked all the alternatives that are available, but the researcher can only observe a few of all the ranks that this individual has assigned. To preserve the order of the observed alternatives and construct a new choice set for the individual without making any assumptions about the outside options, I construct a new rank-ordered list using only the alternatives with observed assigned ranks. Finally, note that the functional form of the candidates' probability of submitting a particular ranked-ordered list (Equation 9) would eliminate individual fixed effects by construction if they were included in μ_{ij}.

To consistently estimate causal effects under the previously stated assumptions, one needs to address the endogeneity of the max-matching probability and the selection issue of the matched candidates' observed ranked-ordered lists. Regarding the endogeneity issue, the control function is added to the systematic component of the utility function:

$$\tilde{\mu}_{ij} = \beta^w \ln w_j + \beta^p p_{ij} + \beta^q q_j + \beta^X X_j + \lambda_p \hat{v}_{ij}^p \tag{10}$$

The coefficients of the systematic component are estimated via log-likelihood maximization, where the log-likelihood function is:

$$\mathcal{L} = \sum_{i=1}^{I} \sum_{j=1}^{J} R_{ij} \tilde{\mu}_{ij} - \sum_{i=1}^{I} \sum_{j=1}^{J} R_{ij} \ \ln \ \left(\sum_{k=1}^{J} \delta_{ijk} R_{ik} \exp(\tilde{\mu}_{ik}) \right) \tag{11}$$

I restrict the sample used to perform the estimation of Equation 11 to only the waitlisted candidates, with the aim of eliminating the possibility of a selection bias generated by the matched candidates. Then, using the estimated parameters of Equations 7 and 11, I adopt the same willingness to pay (WTP) measure derived in Maestas et al. (2023) to obtain the WTP for each attribute

$\ell \in \{p_{ij}, q_j, \{X_j^n\}_{n=1}^N\}$:

$$\beta^w \ln(w_i) = \beta^\ell + \beta^w[\ln(w_i) - WTP_j^\ell]$$
$$WTP_j^\ell = w_j \left[1 - \exp\left(-\frac{\beta^\ell}{\beta^w}\right)\right] \tag{12}$$

Where $[1 - \exp(-\frac{\beta^\ell}{\beta^w})]$ indicates, on average, the percentage of the monthly earnings that a candidate is willing to pay for a certain attribute of the job posting. Lastly, in Appendix F, I present evidence that the WTP estimates are unbiased by showing the results of a simulation procedure that implements the censored ranked-ordered logit proposed above.

7 Willingness to Pay Estimates

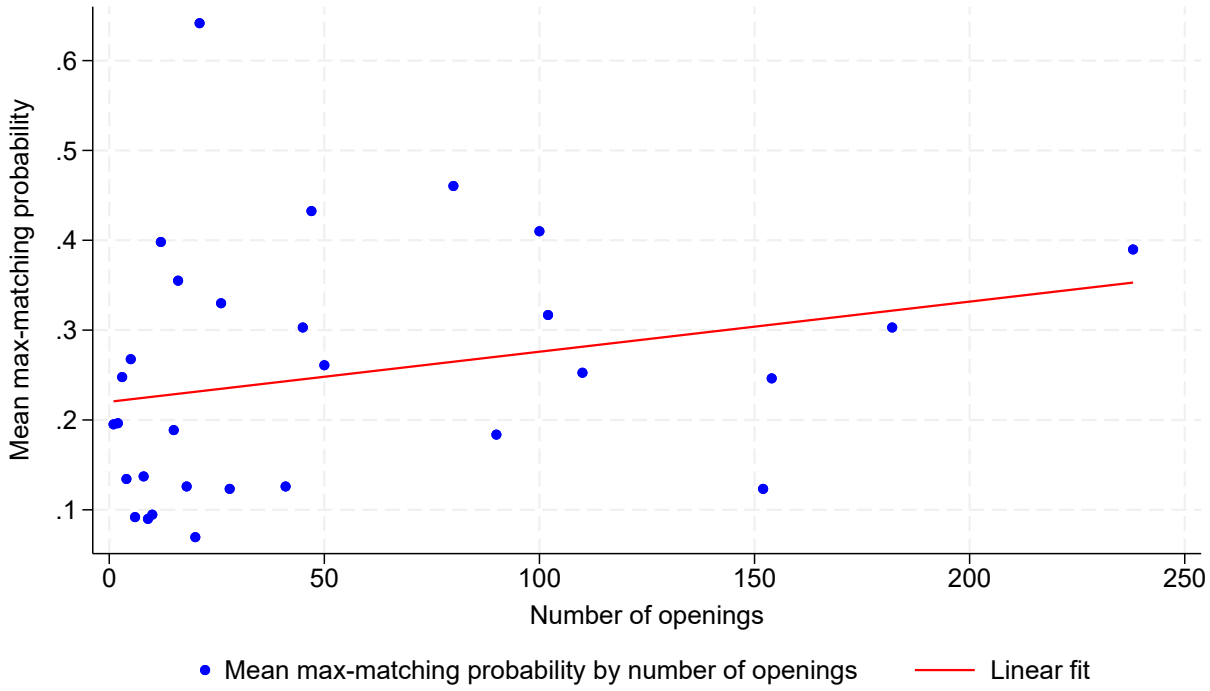
This section shows the willingness to pay estimates derived from the estimation procedures described in Section 6. The main purpose of this section is to identify if the max-matching probability and the number of openings have an impact on the high-performing job candidates' preferences for the job postings and quantify these impacts. Since both the max-matching probability and the number of openings can be interpreted as measures of how competitive it is to get into a particular job, it is important to understand the relationship between the two. Figure 2 illustrates the relationship between the max-matching probability and the number of openings by showing a positive correlation between the mean value of the max-matching probability of the candidates in the full sample for each particular number of openings of the job postings and the number of openings. This positive association pattern is consistent with the conjecture that higher values of both variables indicate a more accessible job.

Next, I separately identify the effect of the max-matching probability and the number of openings, as well as other non-monetary job characteristics, on candidates' preferences for the job postings. Tables 2 and 3 shows which percentage of the earnings provided by a job posting a high-performing candidate is willing to pay, on average, to obtain certain job attributes. In both tables, columns (4) - (6) include the control function derived from the first-stage residual of the max-matching probability in the systematic component of the utility function in the estimation procedure of the model used to estimate the preference parameters[21], while columns (1) - (3) do not include it. The WTP estimates are computed according to Equation 12.

I begin by presenting the results from Table 2, which displays the willingness to pay estimates derived from the control function logit model (Equation 7), where the dependent variable is an indicator variable assigned value 1 if the job posting is ranked as the first option and zero otherwise. In this setting, across all specifications and for all sets of non-monetary job characteristics, the candidates are willing, on average, to trade-off a percentage of the earnings provided by a job posting to obtain a higher max-matching probability, a larger number of openings, many cities

[21]The results for the first-stage regression that provided the residuals used in the estimations of this section are in column (1) of Table 14 in Appendix A.

Figure 2: Relationship between max-matching probability and number of openings



Notes: The graph shows, for the full sample of candidates, the mean value of the max-matching probability obtained by the candidates for each different value of number of openings associated with a job posting and the linear fit between the mean value of the max-matching probability and the number of openings.

in a job posting instead of solely the capital or another larger city, and a public body that is a Ministry. In particular, under the IV specification and controlling for the type of occupation, a high-performing candidate is willing to pay, on average, 3.6 percent of the earnings provided by a job posting to increase its max-matching probability by 10 percentage points, 2.3 percent to have 10 additional openings in a job posting, 6.15 percent to have many cities in the description of the job location, and 11.18 percent to have a Ministry as the public body which is offering the job posting. Hence, there is a strong empirical evidence that all four job posting attributes have a positive impact on the high-performing job candidates' preferences for the job postings, at least when they are deciding which alternative is the most preferred.

As mentioned previously, the reduced-form approach omits information on the trade-off between attributes in the less preferred alternatives, which prevents its results from determining which attributes could affect candidates' decisions in a more general sense, not only their first choices. Additionally, the observed rank-ordered lists from the matched candidates are truncated on their matched job posting, which could potentially introduce a selection bias. To address both concerns, I implement the rank-ordered choice model estimation procedure described in Section 6.4 and display its results in Table 3.

Overall, the direction of the effects is the same as the one obtained with the reduced-form approach for all four attributes of interest, after controlling for the type of occupation. However,

the magnitudes of the effects vary. Under the IV specification and controlling for the type of occupation, the rank-ordered approach indicates that a high-performing candidate is willing to pay, on average, 5.7 percent of the earnings provided by a job posting to increase its max-matching probability by 10 percentage points, 0.2 percent to have 10 additional openings in a job posting, 16.51 percent to have many cities in the description of the job location, and 11.25 percent to have a Ministry as the public body which is offering the job posting. Therefore, the WTP estimates for the number of openings are attenuated with the rank-ordered method, whereas the WTP estimates for the max-matching probability are strengthened. Regarding the other two attributes, the WTP to have many cities in the description of the job location increases considerably, and the WTP to have a Ministry as the public body which is offering the job posting remains basically the same in comparison with the estimates from the reduced form approach.

Table 2: Willingness to pay for job postings' attributes - First option logit choice model

	Baseline specification			IV specification		
	(1)	(2)	(3)	(4)	(5)	(6)
Max-matching probability	0.0058	0.0054	0.0048	0.0053	0.0047	0.0036
	(0.0004)	(0.0004)	(0.0005)	(0.0005)	(0.0005)	(0.0007)
Number of openings	0.0020	0.0017	0.0021	0.0020	0.0017	0.0023
	(0.0001)	(0.0001)	(0.0002)	(0.0001)	(0.0001)	(0.0002)
Many cities		0.1136	0.0561		0.1184	0.0615
		(0.0197)	(0.0252)		(0.0202)	(0.0260)
Ministry		0.0573	0.1094		0.0590	0.1118
		(0.0165)	(0.0170)		(0.0166)	(0.0174)
No. Candidates	5,174	5,174	5,174	5,174	5,174	5,174
No. Observations	21,883	21,883	21,883	21,883	21,883	21,883
Occupation control	No	No	Yes	No	No	Yes

Notes: This table contains the willingness to pay estimates of the control function logit model, where the dependent variable is an indicator variable assigned value 1 if the job posting is ranked as the first option and zero otherwise. Columns (4) - (6) include the control function (first-stage residual of the max-matching probability) in the systematic component of the utility function in the estimation procedure, while columns (1) - (3) do not include it. The estimation procedure uses the full sample of candidates. Standard errors, shown in parentheses, are clustered at the candidate level.

Table 3: Willingness to pay for job postings' attributes - Rank-ordered choice model

	Baseline specification			IV specification		
	(1)	(2)	(3)	(4)	(5)	(6)
Max-matching probability	-0.0042	-0.0032	0.0002	-0.0064	-0.0055	0.0057
	(0.0006)	(0.0006)	(0.0004)	(0.0009)	(0.0009)	(0.0008)
Number of openings	0.0018	0.0014	0.0007	0.0019	0.0015	0.0002
	(0.0001)	(0.0001)	(0.0001)	(0.0001)	(0.0001)	(0.0001)
Many cities		0.1454	0.1498		0.1404	0.1651
		(0.0161)	(0.0124)		(0.0162)	(0.0110)
Ministry		0.1436	0.1242		0.1482	0.1125
		(0.0111)	(0.0090)		(0.0113)	(0.0082)
No. Candidates	2,574	2,574	2,574	2,574	2,574	2,574
No. Observations	14,959	14,959	14,959	14,959	14,959	14,959
Occupation control	No	No	Yes	No	No	Yes

Notes: This table contains the willingness to pay estimates of the rank-ordered choice model, where the dependent variable is a categorical variable indicating the order of the observed rankings submitted by the candidates in their ranked-ordered lists. Columns (4) - (6) include the control function (first-stage residual of the max-matching probability) in the systematic component of the utility function in the estimation procedure, while columns (1) - (3) do not include it. The estimation procedure uses the sample of waitlisted candidates from whom I can observe two or more ranked alternatives in their submitted ranked-ordered lists. Standard errors, shown in parentheses, are clustered at the candidate level.

8 Robustness Checks

The willingness to pay estimates presented in Section 7 do not explore in detail two aspects of the data analyzed in this paper: the problems posed by the use of rank-ordered data and the heterogeneity in candidates' types. This section investigates how these aspects can influence the WTP estimates and develops methods to accommodate them in the estimation procedures.

8.1 Problems with rank-ordered data

The usage of rank-ordered data poses its own challenges. The three most relevant challenges for the estimation procedure that I implemented above are (1) noisy bottom ordering, (2) ties between options, and (3) partially observed choice sets. In the following paragraphs, I will explain how each of these three aspects of rank-ordered data can potentially distort the estimation results.

As noticed in earlier studies (Chapman and Staelin (1982) and Hausman and Ruud (1987)), the individuals might pay less attention to the ordering of least preferred options, making their ranks noisier. An alternative is to only focus on the top choices, as in Ainsworth et al. (2023). They argue

that in their setting most of the respondents have settled on the two first ranked options. However, by opting to only consider the top choices and disregarding the other ones, one could be potentially losing a lot of information about the individuals' preferences in many other settings, where the researcher has no reason to believe that if offered one of the lower-ranked options, the respondent would decline. Table 15 presents the willingness to pay estimates for the IV specification of the rank-ordered choice model, where each column represents a level of truncation on the rank-ordered lists submitted by the candidates[22]. In general, the results are robust for truncation levels of the top 15 choices and above. When the truncation becomes more restrictive, the estimates for the max-matching probability and for the ministry coefficients shrink. One possible explanation for the shrinkage is the lack of variation for these variables within ranked job postings when I impose more extreme sample restrictions, after controlling for other job postings' attributes.

Table 4: Willingness to pay for job postings' attributes - Rank-ordered choice model with distinct bottom truncation levels

	Top 10	Top 15	Top 20	No restriction
Max-matching probability	0.0025	0.0047	0.0058	0.0057
	(0.0014)	(0.0010)	(0.0009)	(0.0008)
Number of openings	0.0006	0.0002	0.0002	0.0002
	(0.0001)	(0.0001)	(0.0001)	(0.0001)
Many cities	0.1452	0.1624	0.1695	0.1651
	(0.0183)	(0.0130)	(0.0116)	(0.0110)
Ministry	0.0270	0.0877	0.1131	0.1125
	(0.0133)	(0.0093)	(0.0085)	(0.0082)
No. Candidates	2,219	2,497	2,561	2,574
No. Observations	9,203	13,440	14,679	14,959
Occupation control	Yes	Yes	Yes	Yes

Notes: This table shows the willingness to pay estimates of the control function logit model, where the dependent variable is an indicator variable assigned value 1 if the job posting is ranked as the first option and zero otherwise. Each column imposes a distinct restriction on the values of the dependent variable, indicated by the column's header. Top X indicates that all values larger than X (alternatives that are less preferred than the X first choices) where dropped from the sample used in the estimation procedure associated with each column. The estimation procedure for all columns uses the full sample of candidates and then restricts the bottom part of their submitted rank-ordered lists. Standard errors, shown in parentheses, are clustered at the candidate level.

Another source of estimation challenges is imposed when there are ties between alternatives. This problem occurs in many surveys where the respondent gives the same numerical ranking to a subset of the options. The most standard way of dealing with ties in ranks is to assume that the individual submitting the rank-ordered list with ties actually has a strict preference order for the

[22]The willingness to pay estimates for the IV specification of the logit model for the same levels of truncation are in Table 4 in the Appendix.

tied options, but the researcher cannot observe it (Allison and Christakis (1994)). Fortunately, I don't have to make any assumptions about ties, because in the assignment mechanism implemented by the government for the first edition of the CPNU, the candidates were not allowed to submit rank-ordered lists with ties.

Lastly, the presence of partially observed choice sets can also potentially introduce biases in the estimation results. This occurs because the researcher cannot observe all the alternatives in each individual's choice set. Even in settings where the researcher can observe all the submitted ranked options by each individual, the lists might not contain all the alternatives in their choice set. There are many reasons why a candidate might not rank all the alternatives in their choice set: Inattention (Hausman and Ruud (1987)), fatigue (Chen and Kesten (2019)), search costs (Arteaga et al. (2022)), to cite a few. However, as discussed in Section 3, the significant amount of time that the candidates had to make their decisions, the reasonably low number of alternatives, the high-stakes environment, and the government guidance on the application procedures mitigate the problems that could arise due to those reasons. Moreover, the censored nature of the rank-ordered lists, which encompasses only the alternatives to which the candidates were either matched or waitlisted, does not seem to inflict any biases in the estimations, as the simulations in Appendix F indicate.

8.2 Heterogeneity in candidates' types

8.2.1 Max-matching probability

I extend the homogeneous model of max-matching probability by relaxing two of its assumptions: (1) all candidates are eligible for all job openings and (2) there is only one type of opening for each job posting. As a consequence, the heterogeneous model allows for heterogeneous eligibility for reserved openings and for the existence of multiple types of openings for each job posting. Assume there are E types of openings indexed from 1 to E for each job posting j, and that each type of opening e is reserved to the candidates with a particular personal observable characteristics (e.g. demographics). The number of openings available in job posting j is described by the vector $q_j = (q_j^1, \cdots, q_j^E) \in \mathbb{Z}_{\geq 0}^E \backslash \{\mathbf{0}\}$. Let $\mathcal{I}_e \subseteq \mathcal{I}$ be the subset of candidates who exhibit the characteristic related to the type of opening e and therefore are eligible to this type of opening for every job posting. Assume that every candidate exhibits characteristic $e = 1$, define $\mathcal{E}(i)$ as the set of personal observable characteristic that candidate i possesses, such that $\mathcal{E}(i) = \{e : i \in \mathcal{I}_e \text{ for all } \mathcal{I}_e \subseteq \mathcal{I}\}$. To partition the candidates in non-overlapping sets, define the candidate type $e^*(i)$ as the maximum index value of a characteristic they possess, $e^*(i) = \max e$ s.t. $e \in \mathcal{E}(i)$[23]. The heterogeneous model of random-effects latent ability is:

$$s_i^k = \delta_{e^*(i)}^k + \lambda_k a_i + \varepsilon_i^k \tag{13}$$

[23]Candidates are allowed to have more than one of the characteristics, but they can only have one type. In other words, the set of candidates can be partitioned in non-overlapping sets of candidates types.

Under the heterogeneous setting, there is an expected value $\delta^k_{e^*(i)}$ for each type of candidate $e^*(i)$ within each component k, the variance of the latent ability differs for each candidate type $e^*(i)$, $a_i \sim \mathcal{N}(0, \gamma^2_{e^*(i)})$, the variance of the error-term σ^2_{gk} varies across both components and candidate types such that $\epsilon_{ik} \sim \mathcal{N}(0, \sigma^2_{e^*(i),k})$, and there is a cutoff for every pair of job posting and type of candidate, c^e_j. The other variables of the model remain the same as in the homogeneous characterization. Similarly to the model presented previously, the candidates' final scores for each job posting has the following distribution:

$$s_{ij} = \sum_{k=1}^{K} \pi_{k,j} s^k_i + \left(1 - \sum_{k=1}^{K} \pi_{k,j}\right) t_{ij} \sim \mathcal{N}\left(\boldsymbol{\pi}_j^\top \boldsymbol{\delta}_{e^*(i)} + (1 - \mathbf{1}^\top \boldsymbol{\pi}_j) t_{ij}, \boldsymbol{\pi}_j^\top \boldsymbol{\Sigma}_{e^*(i)} \boldsymbol{\pi}_j\right) \tag{14}$$

Analogously, the max-matching probability for candidate i in job posting j under heterogeneous distribution of latent ability is:

$$\widetilde{p}_{ij} = \Pr\left(s_{ij} - \left(\min_{e \in \mathcal{E}(i)} c^e_j\right) \geq 0\right) \tag{15}$$

In the estimation procedure of the heterogeneous model, I also normalize $\lambda_1 = 1$ and estimate the the final score parameters $\boldsymbol{\theta} = (\{\delta^1_g, \delta^2_g, \delta^3_g\}^3_{g=1}, \{\gamma^2_g\}^3_{g=1}, \lambda_2, \lambda_3, \{\sigma^2_{g1}, \sigma^2_{g2}, \sigma^2_{g3}\}^3_{g=1})$ using a constrained MLE[24]. Under this framework, the candidate's i plug-in estimator for the max-matching probability for job posting j is:

$$\widehat{\widetilde{p}}^{HET}_{ij} = 1 - \Phi\left(\frac{\left(\min_{e \in \mathcal{E}(i)} c^e_j\right) - \boldsymbol{\pi}_j^\top \boldsymbol{\delta}_{e^*(i)} - (1 - \mathbf{1}^\top \boldsymbol{\pi}_j) t_{ij}}{\sqrt{\boldsymbol{\pi}_j^\top \boldsymbol{\Sigma}_{e^*(i)} \boldsymbol{\pi}_j}}\right) \tag{16}$$

where $e^*(i)$ is candidate's i type (general competition, black person, person with disability)[25], $\mathcal{E}(i)$ is a set that included general competition for all candidates and might also include black person and/or person with disabilities if the candidate has self-declared these personal characteristics, c^e_j is the cutoff score corresponding to each one of the three types of openings for job posting j, $\boldsymbol{\pi}_j$ and t_{ij} are the same as in the homogeneous model, and $\boldsymbol{\delta}_{e^*(i)}$ and $\boldsymbol{\Sigma}_{e^*(i)}$ are estimated with the components' scores of every candidate from the same type in the full sample through a constrained MLE.

To accommodate the heterogeneity in the max-matching probability, for every candidate, I compute a leave-one-out cutoff score associated with the candidate's type:

$$Z^{HET}_{p_{ij}} = \mathbb{E}[s_{ij}] - c^{e^*(i),IV}_j = \boldsymbol{\pi}_j^\top \hat{\boldsymbol{\delta}}_{\boldsymbol{e^*(i)}} + (1 - \mathbf{1}^\top \boldsymbol{\pi}_j) t_{ij} - c^{e^*(i),IV}_j \tag{17}$$

The number of openings is also adjusted. For each candidate i, the number of openings to

[24] See Appendix E.2 for details.

[25] In this estimation procedure, if the candidate in the sample is self-declared both a black person and a person with disabilities, their type is person with disability. If the candidate is self-declared neither a black person nor a person with disabilities, their type is general competition.

which they are eligible in job posting j is $q_j^{\mathcal{E}(i)}$. It includes the general competition openings and the reserved openings corresponding to their other self-declared demographic characteristics, if there are any. The residual for the control function approach comes from the following first-stage regression of the matching probability p_{ij}^{HET}:

$$p_{ij}^{HET} = b^w \ln w_j + b^Z Z_{p_{ij}}^{HET} + b^q q_j^{\mathcal{E}(i)} + b^X X_j + v_{ij}^p \tag{18}$$

8.2.2 Willingness to pay estimates

This section presents the willingness to pay estimates for job postings' attributes when there are heterogeneous candidate types and openings reserved for a subset of candidates with certain observable demographic characteristics. The main changes with respect to the willingness to pay estimates derived from the homogeneous model are in (1) the modeling and estimation of the max-matching probability, (2) the use of the number of openings that are *eligible* to each candidate, which depends on their observable demographic characteristics, and (3) the estimation of the residual function to control for the endogeneity in the max-matching probability using the new max-matching probability and number of openings variables. The modeling and estimation procedures for these changes are described in Section 8.2.1. After implementing the aforementioned changes, the estimation procedures for the control function logit model and for the rank-ordered choice model are the same as those specified in Sections 6.3 and 6.4.

The homogeneous and the heterogeneous settings are qualitatively similar, where almost every WTP estimate of the four attributes of interest demonstrates that they have a positive impact on candidates' preferences. In addition, the level of the WTP estimates for a job posting that is offered by a Ministry, after controlling for occupation, is fairly stable across all settings, specifications, and choice models, ranging from 10.44 to 12.42 percent, on average, of the job posting's earnings. Moreover, in the rank-ordered choice model under the IV specification and with occupational controls, the WTP to increase the max-matching probability by 10 percentage points is very similar in both the homogeneous and heterogeneous settings, being 5.7 and 5.2 percent, respectively. However, there are a few differences in the results that are worth noticing. The first is that the estimate for the max-matching probability in the heterogeneous setting under the IV specification and with occupational controls derived from the logit choice model is not statistically significant (see column (6) in Table 5). The other is the discrepancy in the level of the WTP estimates for the number of openings provided by the rank-ordered choice model. In the homogeneous setting, under the IV specification and with occupational controls, a high-performing candidate is willing to pay, on average, 0.2 percent of the job posting's earnings to have 10 additional total openings in a job posting, whereas in the heterogeneous setting, they are willing to pay, on average, 0.5 percent of the job posting's earnings to have 10 additional eligible openings.

Table 5: Willingness to pay for job postings' attributes - First option logit choice model with heterogeneity in candidates' types

	Baseline specification			IV specification		
	(1)	(2)	(3)	(4)	(5)	(6)
Max-matching probability	0.0047	0.0044	0.0040	0.0052	0.0050	-0.0001
	(0.0003)	(0.0003)	(0.0004)	(0.0009)	(0.0008)	(0.0013)
Number of openings (eligible)	0.0024	0.0021	0.0026	0.0025	0.0021	0.0033
	(0.0001)	(0.0001)	(0.0002)	(0.0001)	(0.0002)	(0.0003)
Many cities		0.1108	0.0579		0.1053	0.0461
		(0.0193)	(0.0243)		(0.0188)	(0.0278)
Ministry		0.0518	0.1044		0.0807	0.1227
		(0.0165)	(0.0171)		(0.0148)	(0.0184)
No. Candidates	5,174	5,174	5,174	5,174	5,174	5,174
No. Observations	21,883	21,883	21,883	21,883	21,883	21,883
Occupation control	No	No	Yes	No	No	Yes

Notes: This table contains the willingness to pay estimates of the control function logit model, where the dependent variable is an indicator variable assigned value 1 if the job posting is ranked as the first option and zero otherwise. Columns (4) - (6) include the control function (first-stage residual of the max-matching probability) in the systematic component of the utility function in the estimation procedure, while columns (1) - (3) do not include it. The estimation procedure uses the full sample of candidates. Standard errors, shown in parentheses, are clustered at the candidate level.

Table 6: Willingness to pay for job postings' attributes - Rank-ordered choice model with heterogeneity in candidates' types

	Baseline specification			IV specification		
	(1)	(2)	(3)	(4)	(5)	(6)
Max-matching probability	-0.0034	-0.0028	-0.0001	-0.0024	0.0007	0.0052
	(0.0005)	(0.0005)	(0.0003)	(0.0012)	(0.0010)	(0.0007)
Number of openings (eligible)	0.0023	0.0018	0.0009	0.0020	0.0014	0.0005
	(0.0002)	(0.0001)	(0.0001)	(0.0001)	(0.0001)	(0.0001)
Many cities		0.1454	0.1481		0.1387	0.1495
		(0.0163)	(0.0125)		(0.0149)	(0.0106)
Ministry		0.1455	0.1242		0.1437	0.1216
		(0.0111)	(0.0090)		(0.0101)	(0.0079)
No. Candidates	2,574	2,574	2,574	2,574	2,574	2,574
No. Observations	14,959	14,959	14,959	14,959	14,959	14,959
Occupation control	No	No	Yes	No	No	Yes

Notes: This table contains the willingness to pay estimates of the rank-ordered choice model, where the dependent variable is a categorical variable indicating the order of the observed rankings submitted by the candidates in their ranked-ordered lists. Columns (4) - (6) include the control function (first-stage residual of the max-matching probability) in the systematic component of the utility function in the estimation procedure, while columns (1) - (3) do not include it. The estimation procedure uses the sample of waitlisted candidates from whom I can observe two or more ranked alternatives in their submitted ranked-ordered lists. Standard errors, shown in parentheses, are clustered at the candidate level.

9 Conclusion

Previous literature shows that job attributes influence candidates' application decisions, particularly non-monetary attributes tied to the job's level of competitiveness. This paper builds on the existing literature by analyzing how high-performing applicants respond to two non-monetary attributes, number of openings and max-matching probability, when they can be simultaneously inferred from job postings. To conduct the analysis, I combine data on individual preferences from applicants' rank-ordered lists and job-attribute data from public bodies' official job descriptions, and compute candidates' willingness to pay for each attribute.

In sum, I find that top-performing candidates prefer job postings that have more openings and that they have a greater chance of being matched with. This indicates that the effects of the two attributes move in the same direction in high-stakes decision-making environments. The findings shed light on a less studied dimension of candidates' decision process: how the perception of competition can influence how candidates evaluate the alternatives they already chose to apply for, not only the decision to apply or not, the latter being the main focus of existing literature in job search.

The results in this study suggest that top-performing job candidates display competition avoidance behavior even when they understand the mechanism, have plenty of time to make their decisions, and the real-world consequences of their choices are high-stakes. This decision-making pattern could be observed even when candidates are not trying to strategically deviate from truth-telling when submitting their preferences and should be taken into account by policymakers who want to implement incentive-compatible matching algorithms that induce truth-telling.

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Appendix

A Tables

Table 7: Total job openings by theme block

1 — Infrastructure, mathematics and engineering	744 job openings
2 — Technology, data and information	580 job openings
3 — Environmental, agrarian and biological	538 job openings
4 — Server work and health	971 job openings
5 — Education, Health, Social Development and Human Rights	1,008 job openings
6 — Economic sectors and regulation	370 job openings
7 — Government management and public administration	1,737 job openings
8 — High School level	692 job openings
Total	6,640 job openings

Source: Brazilian Ministry of Management and Innovation in Public Services, 2024

Table 8: Total job postings by theme block

1 — Infrastructure, mathematics and engineering	32 job postings
2 — Technology, data and information	21 job postings
3 — Environmental, agrarian and biological	21 job postings
4 — Server work and health	9 job postings
5 — Education, Health, Social Development and Human Rights	21 job postings
6 — Economic sectors and regulation	17 job postings
7 — Government management and public administration	43 job postings
8 — High School level	9 job postings
Total	173 job postings

Source: Brazilian Ministry of Management and Innovation in Public Services, 2024.

Table 9: Weighting schemes for candidates' scores

	General	Specific	Discursive	Titles
Weighting scheme I	20	50	20	10
Weighting scheme II	25	50	20	5
Weighting scheme III	25	55	20	0

Notes: This table illustrates the three different weighting schemes used by the job postings in theme block 7 as a priority index in the matching mechanism. Each job posting uses one of these three schemes. The first three rows correspond to the weights given to the exam components, general knowledge, specific knowledge and discursive, in each scheme. The last row corresponds to the weights given to the candidates titles in each scheme. Each row adds up to 100.

Table 10: Job postings - summary statistics

Job posting	Public body	Occupation	Brasilia	Rio de Janeiro	Many cities	Earnings (no titles)	Earnings (with doctorate)	Job openings
1	Attorney General's Office - AGU	Same as Major	No	No	Yes	6761.72	6761.72	154
2	Attorney General's Office - AGU	Analyst	No	No	Yes	6761.72	6761.72	90
3	Attorney General's Office - AGU	Same as Major	No	No	Yes	6761.72	6761.72	2
4	Attorney General's Office - AGU	Same as Major	No	No	Yes	6761.72	6761.72	47
5	Attorney General's Office - AGU	Technician	No	No	Yes	6761.72	6761.72	9
6	National Foundation for Indigenous Peoples - FUNAI	Specialist	No	No	Yes	8263.06	8263.06	26
7	National Foundation for Indigenous Peoples - FUNAI	Specialist	Yes	No	No	8263.06	8263.06	1
8	National Foundation for Indigenous Peoples - FUNAI	Specialist	No	No	Yes	8263.06	8263.06	6
9	National Foundation for Indigenous Peoples - FUNAI	Specialist	No	No	Yes	8263.06	8263.06	10
10	National Foundation for Indigenous Peoples - FUNAI	Specialist	No	No	Yes	8263.06	8263.06	12
11	Brazilian Institute of Geography and Statistics - IBGE	Analyst	No	No	Yes	8453	10547.18	26
12	Brazilian Institute of Geography and Statistics - IBGE	Analyst	No	Yes	No	8453	10547.18	6
13	Brazilian Institute of Geography and Statistics - IBGE	Analyst	No	Yes	No	8453	10547.18	4
14	Brazilian Institute of Geography and Statistics - IBGE	Analyst	No	No	Yes	8453	10547.18	152
15	Brazilian Institute of Geography and Statistics - IBGE	Technician	No	Yes	No	8453	10547.18	2
16	Brazilian Institute of Geography and Statistics - IBGE	Technician	No	Yes	No	8453	10547.18	4
17	Brazilian Institute of Geography and Statistics - IBGE	Technician	No	No	Yes	8453	10547.18	28
18	Brazilian Institute of Geography and Statistics - IBGE	Technician	No	Yes	No	8453	10547.18	2
19	Brazilian Institute of Geography and Statistics - IBGE	Technician	No	No	Yes	8453	10547.18	41
20	Brazilian Institute of Geography and Statistics - IBGE	Technician	No	Yes	No	8453	10547.18	10
21	National Institute for Colonization and Agrarian Reform - INCRA	Analyst	No	No	Yes	5212.29	5212.29	8
22	National Institute for Colonization and Agrarian Reform - INCRA	Analyst	No	No	Yes	5212.29	5212.29	21
23	National Institute for Colonization and Agrarian Reform - INCRA	Analyst	No	No	Yes	5212.29	5212.29	102
24	National Institute for Colonization and Agrarian Reform - INCRA	Analyst	No	No	Yes	5212.29	5212.29	238
25	National Institute of Educational Studies and Research - INEP	Researcher	Yes	No	No	7938.73	7938.73	20
26	Ministry of Agriculture and Livestock - MAPA	Analyst	Yes	No	No	6662.68	6662.68	18
27	Ministry of Science, Technology and Innovation - MCTI	Analyst	No	No	Yes	6662.68	10823.89	3
28	Ministry of Science, Technology and Innovation - MCTI	Analyst	No	No	Yes	6662.68	10823.89	5
29	Ministry of Science, Technology and Innovation - MCTI	Analyst	No	No	Yes	6662.68	10823.89	15
30	Ministry of Science, Technology and Innovation - MCTI	Analyst	No	No	Yes	6662.68	10823.89	5
31	Ministry of Science, Technology and Innovation - MCTI	Analyst	No	No	Yes	6662.68	10823.89	110
32	Ministry of Culture - MinC	Analyst	Yes	No	No	5218.7	5218.7	50
33	Ministry of Management and Innovation in Public Services - MGI	Analyst	No	No	Yes	5218.7	5218.7	182
34	Ministry of Management and Innovation in Public Services - MGI	Same as Major	No	No	Yes	5218.7	5218.7	16
35	Ministry of Management and Innovation in Public Services - MGI	Same as Major	No	No	Yes	5218.7	5218.7	4
36	Ministry of Management and Innovation in Public Services - MGI	Same as Major	Yes	No	No	5218.7	5218.7	5
37	Ministry of Management and Innovation in Public Services - MGI	Specialist	Yes	No	No	20924.8	20924.8	80
38	Ministry of Management and Innovation in Public Services - MGI	Technician	Yes	No	No	5218.7	5218.7	10
39	Ministry of Justice and Public Security - MJSP	Analyst	Yes	No	No	5218.7	5218.7	100
40	Ministry of Health - MS	Technician	Yes	No	No	6662.68	10823.89	9
41	Ministry of Development, Industry, Trade and Services - MDIC	Analyst	Yes	No	No	5218.7	5218.7	50
42	Ministry of Planning and Budget - MPO	Analyst	Yes	No	No	5218.7	5218.7	45
43	National Superintendence of Supplementary Pension Plans - PREVIC	Analyst	Yes	No	No	12953.29	12953.29	9

Notes: This table presents the summary statistics of each job postings in theme block 7. The first column refers to the job postings IDs. The second column shows the public body associated with each job posting. The third column display the broad occupational category corresponding to each job posting. The fourth, fifth, and sixth columns elicit the location of each job posting. The seventh and eighth columns present the earnings of each job posting, for candidates with no advanced degrees other than a college major and for candidates with a doctorate degree, respectively. The last column shows the number of job openings available for each job posting.

Table 11: Job postings - summary statistics per public body

	# of job postings	% Brasilia	% Rio de Janeiro	% Many cities	Mean Earnings (no titles)	Max Earnings (no titles)	Mean Earnings (with doctorate)	Max Earnings (with doctorate)	Mean job openings
Attorney General's Office - AGU	5	0	0	100	6761.72	6761.72	6761.72	6761.72	60.4
Brazilian Institute of Geography and Statistics - IBGE	10	0	60	40	8453	8453	10547.18	10547.18	27.5
Ministry of Agriculture and Livestock - MAPA	1	100	0	0	6662.68	6662.68	6662.68	6662.68	18
Ministry of Culture - MinC	1	100	0	0	5218.7	5218.7	5218.7	5218.7	50
Ministry of Development, Industry, Trade and Services - MDIC	1	100	0	0	5218.7	5218.7	5218.7	5218.7	50
Ministry of Health - MS	1	100	0	0	6662.68	6662.68	10823.89	10823.89	9
Ministry of Justice and Public Security - MJSP	1	100	0	0	5218.7	5218.7	5218.7	5218.7	100
Ministry of Management and Innovation in Public Services - MGI	6	50	0	50	7836.38	20924.8	7836.38	20924.8	49.5
Ministry of Planning and Budget - MPO	1	100	0	0	5218.7	5218.7	5218.7	5218.7	45
Ministry of Science, Technology and Innovation - MCTI	5	0	0	100	6662.68	6662.68	10823.89	10823.89	27.6
National Foundation for Indigenous Peoples - FUNAI	5	20	0	80	8263.06	8263.06	8263.06	8263.06	11
National Institute for Colonization and Agrarian Reform - INCRA	4	0	0	100	5212.29	5212.29	5212.29	5212.29	92.25
National Institute of Educational Studies and Research - INEP	1	100	0	0	7938.73	7938.73	7938.73	7938.73	20
National Superintendence of Supplementary Pension Plans - PREVIC	1	100	0	0	12953.29	12953.29	12953.29	12953.29	9

Notes: This table presents the summary statistics of each public body associated with at least one job posting in theme block 7. The first column indicates how many job posting are associated with each public body. The second, third, and fourth columns add up to 100 and display the percentage of job postings corresponding to each public body which are located in the location associated with one of the three columns. The fifth and the seventh columns present the mean earnings of job postings in public bodies associated with each row for candidates with no advanced degrees other than a college major and for candidates with a doctorate degree, respectively. The sixth and the eighth columns present the maximum earnings of job postings in public bodies associated with each row for candidates with no advanced degrees other than a college major and for candidates with a doctorate degree, respectively. The last column shows the mean number of job openings per job posting for each public body.

Table 12: Number of openings - summary statistics

Job posting ID	Total	General competition (% of Total)	Black (% of Total)	Person with Disabilities (% of Total)	Indigenous (% of Total)
1	154	74.7	20.1	5.2	0.0
2	90	74.4	20.0	5.6	0.0
3	2	100.0	0.0	0.0	0.0
4	47	74.5	19.1	6.4	0.0
5	9	66.7	22.2	11.1	0.0
6	26	42.3	19.2	7.7	30.8
7	1	100.0	0.0	0.0	0.0
8	6	66.7	16.7	16.7	0.0
9	10	40.0	20.0	10.0	30.0
10	12	41.7	16.7	8.3	33.3
11	26	73.1	19.2	7.7	0.0
12	6	66.7	16.7	16.7	0.0
13	4	75.0	25.0	0.0	0.0
14	152	75.0	19.7	5.3	0.0
15	2	100.0	0.0	0.0	0.0
16	4	75.0	25.0	0.0	0.0
17	28	71.4	21.4	7.1	0.0
18	2	100.0	0.0	0.0	0.0
19	41	73.2	19.5	7.3	0.0
20	10	70.0	20.0	10.0	0.0
21	8	62.5	25.0	12.5	0.0
22	21	71.4	19.0	9.5	0.0
23	102	74.5	19.6	5.9	0.0
24	238	74.8	20.2	5.0	0.0
25	20	75.0	20.0	5.0	0.0
26	18	72.2	22.2	5.6	0.0
27	3	66.7	33.3	0.0	0.0
28	5	60.0	20.0	20.0	0.0
29	15	73.3	20.0	6.7	0.0
30	5	60.0	20.0	20.0	0.0
31	110	74.5	20.0	5.5	0.0
32	50	74.0	20.0	6.0	0.0
33	182	74.7	19.8	5.5	0.0
34	16	75.0	18.8	6.2	0.0
35	4	75.0	25.0	0.0	0.0
36	5	60.0	20.0	20.0	0.0
37	80	75.0	20.0	5.0	0.0
38	10	70.0	20.0	10.0	0.0
39	100	75.0	20.0	5.0	0.0
40	9	66.7	22.2	11.1	0.0
41	50	74.0	20.0	6.0	0.0
42	45	73.3	20.0	6.7	0.0
43	9	66.7	22.2	11.1	0.0
All openings	1737	73.2	19.9	6.0	0.9

Notes: The table shows the summary statistics of the number of openings for each job posting in theme block 7. The first column refers to the job postings IDs. The second column shows the total number of job openings available for each job posting. The third, fourth, fifth, and sixth columns elicit the percentage of openings reserved for each type of candidate in each job posting. The last row shows the total openings and the percentages reserved for each types of candidates.

Table 13: Number of observed ranked alternatives - summary statistics

	Only one observation or matched	Two or more observations & unmatched	Total
All candidates	6,961 (31.8%)	14,959 (68.2%)	21,920 (100.0%)
No affirmative action			
No	1,980 (28.4%)	4,451 (29.8%)	6,431 (29.3%)
Yes	4,981 (71.6%)	10,508 (70.2%)	15,489 (70.7%)
Black			
No	5,425 (77.9%)	11,409 (76.3%)	16,834 (76.8%)
Yes	1,536 (22.1%)	3,550 (23.7%)	5,086 (23.2%)
Person with disability			
No	6,489 (93.2%)	13,906 (93.0%)	20,395 (93.0%)
Yes	472 (6.8%)	1,053 (7.0%)	1,525 (7.0%)
Indigenous			
No	6,924 (99.5%)	14,959 (100.0%)	21,883 (99.8%)
Yes	37 (0.5%)	0 (0.0%)	37 (0.2%)

Notes: The table presents the summary statistics for the number of observed ranked alternatives. For each sample selection, the *Yes* and *No* percentages are with respect to each demographic group.

Table 14: First-stage estimates - control function approach

	(1) Homogeneous model	(2) Heterogeneous model
Instrument	6.959***	
	(0.0868)	
Instrument (heterogeneous)		0.711***
		(0.102)
Number of openings (all)	0.0382***	
	(0.00224)	
Number of openings (eligible)		0.0950***
		(0.00366)
ln(*earnings*)	-12.88***	-24.83***
	(0.408)	(0.892)
Many cities	-4.331***	-4.443***
	(0.254)	(0.419)
Ministry	2.969***	2.674***
	(0.169)	(0.320)
Constant	159.3***	244.0***
	(3.662)	(7.863)
Occupation control	Yes	Yes
Observations	21883	21883
R-squared	0.559	0.335
F-statistic	4112.7	2873.4

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Notes: The table shows, for the full sample of candidates, the first-stage OLS estimates for the control function approach. In column (1), the dependent variable is the max-matching probability estimated using the homogeneous model and the instrument is the the distance between the candidate's score and the leave-one-out general competition cutoff score. In column (2), the dependent variable is the max-matching probability estimated using the heterogeneous model and the instrument is the the distance between the candidate's score and the leave-one-out cutoff score corresponding to the candidate's type. Standard errors, shown in parentheses, are clustered at the candidate level.

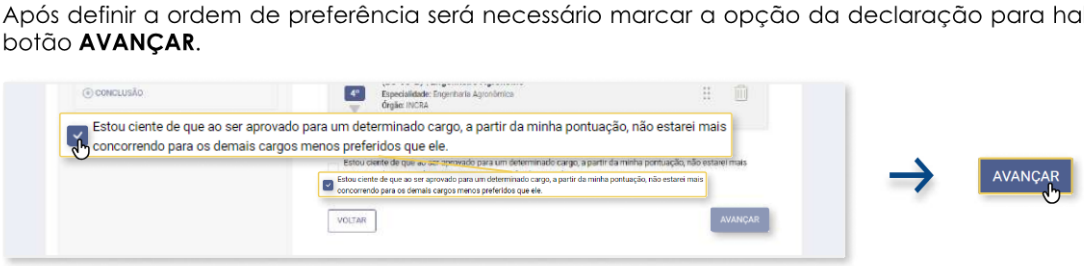
Table 15: Willingness to pay for job postings' attributes - First option logit choice model with distinct bottom truncation levels

	Top 10	Top 15	Top 20	No restriction
Max-matching probability	0.0042	0.0037	0.0036	0.0036
	(0.0009)	(0.0007)	(0.0007)	(0.0007)
Number of openings	0.0028	0.0023	0.0023	0.0023
	(0.0002)	(0.0002)	(0.0002)	(0.0002)
Many cities	-0.0080	0.0519	0.0605	0.0615
	(0.0358)	(0.0286)	(0.0268)	(0.0260)
Ministry	0.1016	0.1041	0.1106	0.1118
	(0.0222)	(0.0189)	(0.0179)	(0.0174)
No. Candidates	4,913	5,125	5,165	5,174
No. Observations	15,643	20,266	21,580	21,883
Occupation control	Yes	Yes	Yes	Yes

Notes: This table shows the willingness to pay estimates of the rank-ordered choice model, where the dependent variable is a categorical variable indicating the order of the observed rankings submitted by the candidates in their ranked-ordered lists. Each column imposes a distinct restriction on the values of the dependent variable, indicated by the column's header. Top X indicates that all values larger than X (alternatives that are less preferred than the X first choices) where dropped from the sample used in the estimation procedure associated with each column. The estimation procedure for all columns uses the sample of waitlisted candidates from whom I can observe two or more ranked alternatives in their submitted ranked-ordered lists. Standard errors, shown in parentheses, are clustered at the candidate level.

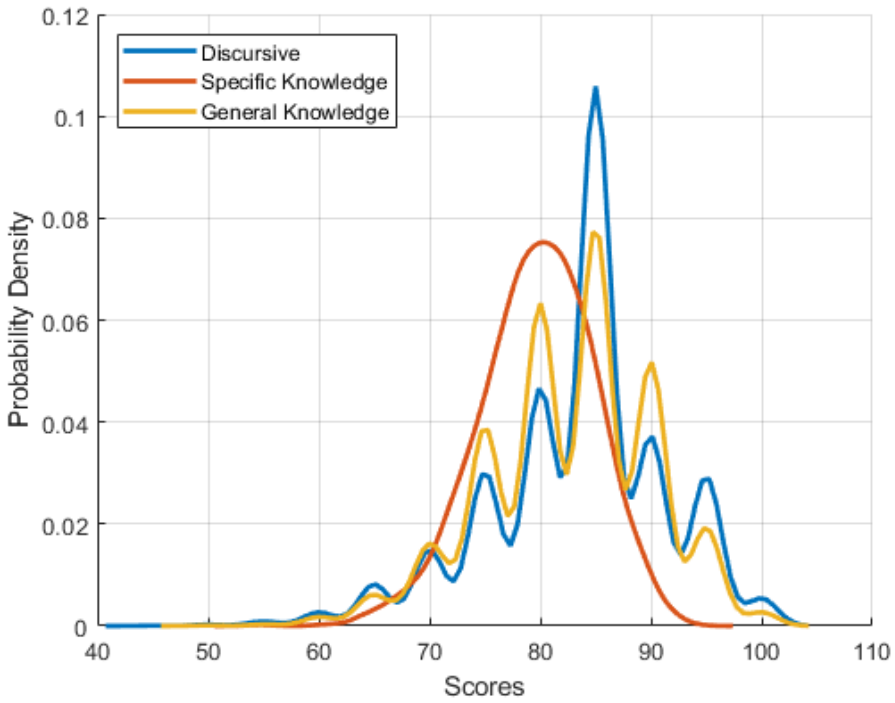
B Figures

Figure 3: Truth-telling attention check



Notes: The English translation for the sentence in the highlighted box is "*I am aware that, upon being approved for a particular position, based on my score, I will no longer be eligible for other, less desirable positions.*"

Figure 4: Distribution of exam components' scores



Notes: The graph shows, for the full sample of candidates, the kernel density estimate of the exam components' scores constructed using the normalized values between 0 and 100.

C Locations

Listed below are all the different locations present in the *edital* description of at least one job opening in theme block 7. Each location is a set that contains at least one city. However, many locations

contain more than one city. When the element is a two-letter state abbreviation, the job applicant can be assigned to any city within the state. When the element is "National", the job applicant can be allocated to any city within Brazil. When the element is "Preferred Capital", the job applicant could be allocated to any Brazilian state capital, but preference to each capital will be given to the applicant that are from its corresponding state.

1. AC; AM; AP; MT; PA; RO; Brasilia DF; Rio de Janeiro RJ
2. AC; AM; MG; MT; PA; PB; SC; Brasilia DF; Rio de Janeiro RJ
3. Brasilia DF
4. Brasilia DF; Rio de Janeiro RJ
5. Brasilia DF; Rio de Janeiro RJ; Cachoeiro do Itapemirim ES; Petropolis RJ; Sao Jose dos Campos SP; Cachoeira Paulista SP; Belem PA; Cuiaba MT; Natal RN; Santa Maria RS; Campinas SP; Santa Teresa ES; Recife PE; Campina Grande PB; Manaus AM; Itajuba MG
6. Brasilia DF; Rio de Janeiro RJ; Sao Jose dos Campos SP; Campinas SP; Santa Teresa ES; Cuiaba MT; Recife PE; Campina Grande PB; Manaus AM; Belem PA
7. Preferred Capital
8. GO; MT; Brasilia DF; Rio de Janeiro RJ
9. Maceio AL; Manaus AM; Macapa AP; Goiania GO; Sao Luis MA; Belo Horizonte MG; Campo Grande MS; Joao Pessoa PB; Recife PE; Curitiba PR; Rio de Janeiro RJ; Boa Vista RR; Porto Alegre RS; Florianopolis SC
10. Maceio AL; Manaus AM; Macapa AP; Salvador BA; Goiania GO; Sao Luis MA; Belo Horizonte MG; Campo Grande MS; Cuiaba MT; Belem PA; Paraiba PB; Teresina PI; Rio de Janeiro RJ; Natal RN; Porto Velho RO; Boa Vista RR; Aracaju SE; Sao Paulo SP; Palmas TO
11. National
12. Rio Branco AC; Maceio AL; Macapa AP; Brasilia DF; Vitoria ES; Goiania GO; Campo Grande MS; Cuiaba MT; Santarem PA; Curitiba PR; Rio de Janeiro RJ; Porto Alegre RS; Sao Paulo SP
13. Rio Branco AC; Maceio AL; Manaus AM; Macapa AP; Salvador BA; Brasilia DF; Vitoria ES; Sao Luis MA; Belo Horizonte MG; Belem PA; Curitiba PR; Rio de Janeiro RJ; Natal RN; Porto Velho RO; Florianopolis SC; Aracaju SE; Sao Paulo SP
14. Rio Branco AC; Maceio AL; Manaus AM; Macapa AP; Salvador BA; Fortaleza CE; Brasilia DF; Vitoria ES; Goiania GO; Sao Luis MA; Belo Horizonte MG; Campo Grande MS; Cuiaba

MT; Belem PA; Santarem PA; Maraba PA; Joao Pessoa PB; Recife PE; Teresina PI; Curitiba PR; Rio de Janeiro RJ; Natal RN; Porto Velho RO; Boa Vista RR; Porto Alegre RS; Florianopolis SC; Aracaju SE; Sao Paulo SP; Palmas TO

15. Rio Branco AC; Manaus AM; Macapa AP; Brasilia DF; Santarem PA; Natal RN; Palmas TO
16. Rio de Janeiro RJ
17. Rio de Janeiro RJ; Campina Grande PB
18. Rio de Janeiro RJ; Petropolis RJ; Sao Jose dos Campos SP
19. Rio de Janeiro RJ; Petropolis RJ; Sao Jose dos Campos SP; Belem PA

D Broad Occupational Categories

Listed below are all the broad occupational categories in theme block 7 and their corresponding job positions (in Portuguese).

- **Analyst**
  1. Analista Administrativo
  2. Analista Técnico Administrativo
  3. Analista de Planejamento, Gestão e Infraestrutura em Informações Geográficas e Estatísticas
  4. Analista em Ciência e Tecnologia
  5. Analista em Reforma e Desenvolvimento Agrário
- **Specialist**
  1. Especialista em Indigenismo
  2. Especialista em Políticas Públicas e Gestão Governamental (EPPGG)
- **Researcher**
  1. Pesquisador-Tecnologista em Informações e Avaliações Educacionais
- **Technician**
  1. Técnico em Comunicação Social
  2. Tecnologista
  3. Técnico em Informações Geográficas e Estatísticas
- **Same as major**

1. Administrador
2. Arquivista
3. Bibliotecário
4. Contador

E Final Score Model Estimation

Disclaimer: I used Claude Sonnet 4.6 between June and July of 2026 to (1) Formulate the Lagrangian with the constraints (2) Take the derivatives (3) Solve the system of equations to find the solution for the estimators of interest (4) Write the 3 previous steps in a latex file (5) Use the solution found for the estimators and the data description previously stated in the prompt to estimate values of the parameters in a Matlab code. In the initial prompt, I included the general description of the model, the distribution of the variables, and the identification assumptions. I reviewed, edited, and take full responsibility for the content in this Appendix section and in the Matlab code.

E.1 Simple Model

E.1.1 Model Setup

Consider the random-effects latent ability model described in the main text:

$$s_i^k = \delta^k + \lambda_k a_i + \varepsilon_i^k \tag{19}$$

Where for each component k, candidate i receives a score s_i^k, δ^k is the expected value of component k's score, a_i is the candidate's latent ability, and ε_i^k is an error-term. Assume that $a_i \sim \mathcal{N}(0, \gamma^2)$, $\lambda_k \in \mathbb{R}$ is a factor loading, $\varepsilon_i^k \sim \mathcal{N}(0, \sigma_k^2)$, and that the latent ability is independent of the error-term and the error-terms are independent across components. Let $\boldsymbol{s}_i = (s_{i1}, \ldots, s_{iK})^\top$ and $\boldsymbol{\delta} = (\delta_1, \ldots, \delta_K)^\top$. By construction, for each individual i, $\boldsymbol{s}_i \sim \mathcal{N}(\boldsymbol{\delta}, \boldsymbol{\Sigma})$, where the covariance matrix $\boldsymbol{\Sigma} \in \mathbb{R}^{K \times K}$ has diagonal entries $\Sigma_{kk} = \lambda_k^2 \gamma^2 + \sigma_k^2$, off-diagonal entries $\Sigma_{kk'} = \lambda_k \lambda_{k'} \gamma^2$, and can be written as $\boldsymbol{\Sigma} = \gamma^2 \boldsymbol{\lambda}\boldsymbol{\lambda}^\top + \boldsymbol{D}$, where $\boldsymbol{\lambda} = (\lambda_1, \ldots, \lambda_K)^\top \in \mathbb{R}^K$ and $\boldsymbol{D} = \mathrm{diag}(\sigma_1^2, \ldots, \sigma_K^2)$.

To make the computations simpler and aligned with the data, assume that $K = 3$. The covariance matrix is represented by

$$\boldsymbol{\Sigma} = \gamma^2 \boldsymbol{\lambda}\boldsymbol{\lambda}^\top + \boldsymbol{D} = \begin{pmatrix} \lambda_1^2\gamma^2 + \sigma_1^2 & \lambda_1\lambda_2\gamma^2 & \lambda_1\lambda_3\gamma^2 \\ \lambda_1\lambda_2\gamma^2 & \lambda_2^2\gamma^2 + \sigma_2^2 & \lambda_2\lambda_3\gamma^2 \\ \lambda_1\lambda_3\gamma^2 & \lambda_2\lambda_3\gamma^2 & \lambda_3^2\gamma^2 + \sigma_3^2 \end{pmatrix},$$

The four parameters in the off-diagonal entries $(\lambda_1, \lambda_2, \lambda_3, \sigma_1^2)$ are only identifiable up to scale. Hence, I normalize $\lambda_1 = 1$ and the unknown parameters $\boldsymbol{\theta} = (\delta^1, \delta^2, \delta^3, \gamma^2, \lambda_2, \lambda_3, \sigma_1^2, \sigma_2^2, \sigma_3^2)$ are identified.

E.1.2 Constrained Maximum Log-Likelihood

The observations $\{\boldsymbol{s}_i\}_{i=1}^I$ are i.i.d. Substituting in the equality constraint for the covariance matrix and setting $\lambda_1 = 1$, the log-likelihood is

$$\ell(\boldsymbol{\theta}) = -\frac{I}{2}\log\det(2\pi\boldsymbol{\Sigma}) - \frac{1}{2}\sum_{i=1}^{I}(\boldsymbol{s}_i - \boldsymbol{\delta})^\top \boldsymbol{\Sigma}^{-1}(\boldsymbol{s}_i - \boldsymbol{\delta}) \tag{20}$$

To obtain the estimates of the unknown parameters of the model, I maximize the log-likelihood subject to positivity constraints on the variance components and impose the constrained structure of Σ. Let $\lambda_0 \geq 0$ be the Lagrange multiplier for $\gamma^2 \geq 0$ and $\lambda_k \geq 0$ be the Lagrange multipliers for $\sigma_k^2 \geq 0$. The Lagrangian is

$$\mathcal{L}(\boldsymbol{\theta}, \boldsymbol{\mu}) = \ell(\boldsymbol{\theta}) + \mu_0\gamma^2 + \sum_{k=1}^{K}\mu_k\sigma_k^2 \tag{21}$$

with KKT conditions $\mu_0 \geq 0,\ \gamma^2 \geq 0,\ \mu_0\gamma^2 = 0$ and $\lambda_k \geq 0,\ \sigma_k^2 \geq 0,\ \lambda_k\sigma_k^2 = 0\ \ \forall k$

In the interior solution (when all variance parameters are strictly positive), every multiplier is zero and the Lagrangian reduces to the log-likelihood. I solve the unconstrained first-order conditions by taking derivatives with respect to $\boldsymbol{\delta}$ and $\boldsymbol{\Sigma}$ and then impose the structure of the covariance matrix $\boldsymbol{\Sigma}$ to obtain the estimates.

Taking the derivative with respect to $\boldsymbol{\delta}$ gives

$$\begin{aligned}\frac{\partial \ell}{\partial \boldsymbol{\delta}} &= \boldsymbol{\Sigma}^{-1}\sum_{i=1}^{I}(\boldsymbol{s}_i - \boldsymbol{\delta}) = \mathbf{0} \\ \hat{\boldsymbol{\delta}} &= \bar{\boldsymbol{s}} := \frac{1}{I}\sum_{i=1}^{I}\boldsymbol{s}_i\end{aligned} \tag{22}$$

Equivalently, $\hat{\delta}^k = \bar{y}_k$ for each k.

Let $\boldsymbol{V} = \sum_{i=1}^{I}(\boldsymbol{s}_i - \hat{\boldsymbol{\delta}})(\boldsymbol{s}_i - \hat{\boldsymbol{\delta}})^\top$. Taking the derivative with respect to $\boldsymbol{\Sigma}$ using the standard matrix identity and the symmetry of Σ (i.e., $\boldsymbol{\Sigma}^{-1} = \boldsymbol{\Sigma}^\top$) gives

$$\begin{aligned}\frac{\partial \ell}{\partial \boldsymbol{\Sigma}} &= -\frac{I}{2}\boldsymbol{\Sigma}^{-1} + \frac{1}{2}\boldsymbol{\Sigma}^{-1}\boldsymbol{V}\,\boldsymbol{\Sigma}^{-1} = \mathbf{0} \\ \hat{\boldsymbol{\Sigma}} &= \frac{\boldsymbol{V}}{I}\end{aligned} \tag{23}$$

Because $\boldsymbol{\Sigma}$ is a function of the scalar parameters, differentiate it via the chain rule. For any scalar parameter θ,

$$\frac{\partial \ell}{\partial \theta} = \operatorname{tr}\left(\frac{\partial \ell}{\partial \boldsymbol{\Sigma}} \cdot \frac{\partial \boldsymbol{\Sigma}}{\partial \theta}\right), \tag{24}$$

where $\frac{\partial \ell}{\partial \boldsymbol{\Sigma}} = -\frac{I}{2}\boldsymbol{\Sigma}^{-1} + \frac{1}{2}\boldsymbol{\Sigma}^{-1}\boldsymbol{V}\boldsymbol{\Sigma}^{-1}$.

Defining $\boldsymbol{R} := \boldsymbol{\Sigma}^{-1}\boldsymbol{V}\boldsymbol{\Sigma}^{-1} - I\boldsymbol{\Sigma}^{-1} = 2\,\partial\ell/\partial\boldsymbol{\Sigma}$, the FOC for any scalar parameter θ in the interior is $\operatorname{tr}(\boldsymbol{R}\,\partial\boldsymbol{\Sigma}/\partial\theta) = 0$.

Compute $\boldsymbol{\Sigma}^{-1}$ directly using the adjugate formula $\boldsymbol{\Sigma}^{-1} = \operatorname{adj}(\boldsymbol{\Sigma})/\det(\boldsymbol{\Sigma})$, where $\operatorname{adj}(\boldsymbol{\Sigma})_{ij} = C_{ji}$ and $C_{ij} = (-1)^{i+j} M_{ij}$ is the (i,j) cofactor (M_{ij} being the (i,j) minor, i.e. the determinant of the 2×2 submatrix obtained by deleting row i and column j).

The determinant of the covariance matrix $\boldsymbol{\Sigma}$ is:

$$\det(\boldsymbol{\Sigma}) = \sigma_1^2\sigma_2^2\sigma_3^2 + \gamma^2(\sigma_2^2\sigma_3^2 + \lambda_2^2\sigma_1^2\sigma_3^2 + \lambda_3^2\sigma_1^2\sigma_2^2)$$

Because $\boldsymbol{\Sigma}$ is symmetric, the cofactor matrix is also symmetric ($C_{ij} = C_{ji}$), so $\operatorname{adj}(\boldsymbol{\Sigma}) = \boldsymbol{C}^\top = \boldsymbol{C}$. The six distinct cofactors are:

$$\begin{aligned}
C_{11} &= \sigma_2^2\sigma_3^2 + \gamma^2(\lambda_2^2\sigma_3^2 + \lambda_3^2\sigma_2^2),\\
C_{22} &= \sigma_1^2\sigma_3^2 + \gamma^2(\sigma_3^2 + \lambda_3^2\sigma_1^2),\\
C_{33} &= \sigma_1^2\sigma_2^2 + \gamma^2(\sigma_2^2 + \lambda_2^2\sigma_1^2),\\
C_{12} &= C_{21} = -\lambda_2\sigma_3^2\gamma^2,\\
C_{13} &= C_{31} = -\lambda_3\sigma_2^2\gamma^2,\\
C_{23} &= C_{32} = -\lambda_2\lambda_3\sigma_1^2\gamma^2
\end{aligned}$$

Since $\operatorname{adj}(\boldsymbol{\Sigma}) = \boldsymbol{C}$:

$$\boldsymbol{\Sigma}^{-1} = \frac{1}{\det \boldsymbol{\Sigma}} \begin{pmatrix} C_{11} & C_{12} & C_{13} \\ C_{12} & C_{22} & C_{23} \\ C_{13} & C_{23} & C_{33} \end{pmatrix} \tag{25}$$

A key simplification for the FOC derivations below is obtained multiplying $\boldsymbol{\Sigma}^{-1}$ by $\boldsymbol{\lambda}$, which cancels the terms multiplied by γ^2 that are not in the determinant:

$$\boldsymbol{\Sigma}^{-1}\boldsymbol{\lambda} = \frac{1}{\det \boldsymbol{\Sigma}} \begin{pmatrix} \sigma_2^2\sigma_3^2 \\ \lambda_2\,\sigma_1^2\sigma_3^2 \\ \lambda_3\,\sigma_1^2\sigma_2^2 \end{pmatrix} \tag{26}$$

Since $\partial\boldsymbol{\Sigma}/\partial\gamma^2 = \boldsymbol{\lambda}\boldsymbol{\lambda}^\top$, the chain rule (24) gives

$$\frac{\partial \ell}{\partial \gamma^2} = \frac{1}{2}\operatorname{tr}(\boldsymbol{R}\,\boldsymbol{\lambda}\boldsymbol{\lambda}^\top) = \boldsymbol{\lambda}^\top \boldsymbol{R}\boldsymbol{\lambda} = 0 \tag{27}$$

Let $\hat{\boldsymbol{V}} = \boldsymbol{V}/I$ and $\boldsymbol{w} = \boldsymbol{\Sigma}^{-1}\boldsymbol{\lambda}$. Then equation 27 simplifies to

$$\boldsymbol{w}^\top \hat{\boldsymbol{V}}\, \boldsymbol{w} = \boldsymbol{\lambda}^\top \boldsymbol{\Sigma}^{-1}\boldsymbol{\lambda}. \tag{28}$$

Since $\partial\boldsymbol{\Sigma}/\partial\sigma_k^2 = \boldsymbol{e}_k\boldsymbol{e}_k^\top$, the chain rule (24) selects the (k,k) entry of $\boldsymbol{R}$:

$$\frac{\partial \ell}{\partial \sigma_k^2} = \frac{1}{2}R_{kk} = \frac{1}{2}\left[(\boldsymbol{\Sigma}^{-1}\hat{\boldsymbol{V}}\boldsymbol{\Sigma}^{-1})_{kk} - (\boldsymbol{\Sigma}^{-1})_{kk}\right] = 0$$
$$(\boldsymbol{\Sigma}^{-1})_{kk} = (\boldsymbol{\Sigma}^{-1}\hat{\boldsymbol{V}}\boldsymbol{\Sigma}^{-1})_{kk}$$

Let $\boldsymbol{u}_k = \boldsymbol{\Sigma}^{-1}\boldsymbol{e}_k$ denote the k-th column of $\boldsymbol{\Sigma}^{-1}$. Then for each $k = 1, 2, 3$:

$$\boldsymbol{u}_k^\top \hat{\boldsymbol{V}}\, \boldsymbol{u}_k = (\boldsymbol{\Sigma}^{-1})_{kk} \tag{29}$$

Finally, since $\partial\boldsymbol{\Sigma}/\partial\lambda_k = \gamma^2(\boldsymbol{e}_k\boldsymbol{\lambda}^\top + \boldsymbol{\lambda}\boldsymbol{e}_k^\top)$ for $k = 2, 3$:

$$\operatorname{tr}\Big(\boldsymbol{R}\,\gamma^2(\boldsymbol{e}_2\boldsymbol{\lambda}^\top + \boldsymbol{\lambda}\boldsymbol{e}_2^\top)\Big) = 2\gamma^2\,\boldsymbol{e}_k^\top\boldsymbol{R}\boldsymbol{\lambda} = 0$$

Assuming an interior solution where $\gamma^2 > 0$, it gives

$$\boldsymbol{e}_k^\top\boldsymbol{\Sigma}^{-1}\hat{\boldsymbol{V}}\,\boldsymbol{v} = \boldsymbol{e}_k^\top\boldsymbol{\Sigma}^{-1}\boldsymbol{\lambda} = \frac{\lambda_k\sigma_1^2\sigma_3^2}{\det\boldsymbol{\Sigma}} \tag{30}$$

Equations (28)–(30) form a system of six nonlinear equations in six unknowns $(\gamma^2, \lambda_2, \lambda_3, \sigma_1^2, \sigma_2^2, \sigma_3^2)$, which I solve numerically with a Newton–Raphson algorithm. A closed-form solution is not available in general because $\boldsymbol{\Sigma}^{-1}$ depends nonlinearly and simultaneously on all six parameters. A natural starting point for the Newton–Raphson algorithm is to match the three theoretical off-diagonal entries $\Sigma_{12} = \lambda_2\gamma^2$, $\Sigma_{13} = \lambda_3\gamma^2$, $\Sigma_{23} = \lambda_2\lambda_3\gamma^2$ to their sample counterparts $\hat{\gamma}_0^2 = \frac{\hat{V}_{12}\hat{V}_{13}}{\hat{V}_{23}}$, $\hat{\lambda}_{2,0} = \frac{\hat{V}_{23}}{\hat{V}_{13}}$, $\hat{\lambda}_{3,0} = \frac{\hat{V}_{23}}{\hat{V}_{12}}$, and $\hat{\sigma}_{k,0}^2 = \max\Big(0,\, \hat{V}_{kk} - \hat{\lambda}_{k,0}^2\hat{\gamma}_0^2\Big)$.

E.2 Heterogeneous Model

E.2.1 Model Setup

The individuals are now partitioned into G subgroups $\mathcal{G}_1, \ldots, \mathcal{G}_G$ of sizes $I_1, \ldots, I_G$ such that $\sum_g I_g = I$. The random-effects latent ability model becomes:

$$s_i^k = \delta_g^k + \lambda_k a_i + \varepsilon_i^k \tag{31}$$

where the variance of the error-term σ_{gk}^2 varies across both components and groups, while the variance of the candidates' latent ability varies within group g and is the same across components for the same group, such that $\epsilon_{ik} \sim \mathcal{N}(0, \sigma_{gk}^2)$ and $a_i \sim \mathcal{N}(0, \gamma_g^2)$. Assuming that the latent ability is independent of the error-term and the error-terms are independent across components, for each individual $i \in \mathcal{G}_g$, $\boldsymbol{s}_i \sim \mathcal{N}(\boldsymbol{\delta}_g, \boldsymbol{\Sigma}_g)$ and the covariance matrix can be written as $\boldsymbol{\Sigma}_g = \gamma_g^2\boldsymbol{\lambda}\boldsymbol{\lambda}^\top + \boldsymbol{D}_g$. Again, assume that $K = 3$ and additionally assume that $G = 3$ and $\lambda_1 = 1$. The unknown parameters are $\boldsymbol{\theta} = (\{\delta_g^1, \delta_g^2, \delta_g^3\}_{g=1}^3,\ \{\gamma_g^2\}_{g=1}^3,\ \lambda_2, \lambda_3, \{\sigma_{g1}^2, \sigma_{g2}^2, \sigma_{g3}^2\}_{g=1}^3)$.

E.2.2 Constrained Maximum Log-Likelihood

The log-likelihood can be written as the sum of the log-likelihoods of each non-overlapping group g

$$\ell(\boldsymbol{\theta}) = \sum_{g=1}^{G} \ell_g = \sum_{g=1}^{G} \left(-\frac{I_g}{2} \log\det(2\pi\boldsymbol{\Sigma}) - \frac{1}{2} \sum_{i \in \mathcal{G}_g} (\boldsymbol{s}_i - \boldsymbol{\delta_g})^\top \boldsymbol{\Sigma}^{-1} (\boldsymbol{s}_i - \boldsymbol{\delta_g}) \right) \tag{32}$$

The Lagrangian is

$$\mathcal{L}(\boldsymbol{\theta}, \boldsymbol{\mu}) = \ell(\boldsymbol{\theta}) + \sum_{g=1}^{G} \mu_g^{(\gamma)} \gamma_g^2 + \sum_{g=1}^{G} \sum_{k=1}^{3} \mu_{gk}^{(\sigma)} \sigma_{gk}^2, \tag{33}$$

with KKT conditions $\mu_g^{(\gamma)} \geq 0$, $\gamma_g^2 \geq 0$, $\mu_g^{(\gamma)} \gamma_g^2 = 0$ for all g, and $\mu_{gk}^{(\sigma)} \geq 0$, $\sigma_{gk}^2 \geq 0$, $\mu_{gk}^{(\sigma)} \sigma_{gk}^2 = 0$ for all (g, k).

Since only the log-likelihood corresponding to group g, ℓ_g, depends on $\boldsymbol{\delta}_g$, the first-order condition with respect to δ_{gk} gives

$$\frac{\partial \mathcal{L}}{\partial \boldsymbol{\delta}_g} = \boldsymbol{\Sigma}_g^{-1} \sum_{i \in \mathcal{G}_g} (\boldsymbol{s}_i - \boldsymbol{\delta}_g) = \mathbf{0} \implies \hat{\delta}_{gk} = \bar{s}_k^{(g)} = \frac{1}{I_g} \sum_{i \in \mathcal{G}_g} s_{ik} \tag{34}$$

Analogously, the same chain rule (24) can be derived for each group g. Define

$$\boldsymbol{V}_g = \sum_{i \in \mathcal{G}_g} (\boldsymbol{s}_i - \boldsymbol{\delta}_g)(\boldsymbol{s}_i - \boldsymbol{\delta}_g)^\top, \qquad \hat{\boldsymbol{V}}_g = \frac{\boldsymbol{V}_g}{I_g}, \qquad \boldsymbol{R}_g := \boldsymbol{\Sigma}_g^{-1} \hat{\boldsymbol{V}}_g \boldsymbol{\Sigma}_g^{-1} - \boldsymbol{\Sigma}_g^{-1}, \qquad \boldsymbol{u}_{g,k} = \boldsymbol{\Sigma}_g^{-1} \boldsymbol{e}_k$$

Since $\partial \boldsymbol{\Sigma}_g / \partial \gamma_g^2 = \boldsymbol{\lambda}\boldsymbol{\lambda}^\top$ and only ℓ_g depends on γ_g^2, the first-order condition with respect to γ_g^2 gives

$$\frac{\partial \mathcal{L}}{\partial \gamma_g^2} = \frac{1}{2} \boldsymbol{\lambda}^\top \boldsymbol{R}_g \boldsymbol{\lambda} + \mu_g^{(\gamma)} = 0.$$

In the interior ($\lambda_g^{(\gamma)} = 0$), setting $\boldsymbol{w}_g = \boldsymbol{\Sigma}_g^{-1} \boldsymbol{\lambda}$:

$$\boldsymbol{w}_g^\top \hat{\boldsymbol{V}}_g \boldsymbol{w}_g = \boldsymbol{\lambda}^\top \boldsymbol{\Sigma}_g^{-1} \boldsymbol{\lambda}, \tag{35}$$

which has the same explicit form as (28) with $\gamma^2 \to \gamma_g^2$ and $\hat{V}_{kk'} \to \hat{V}_{kk'}^{(g)}$. Hence, there are G equations equivalent to (28), one for each group g.

The first-order condition with respect to σ_k^2 gives

$$\frac{\partial \mathcal{L}}{\partial \sigma_k^2} = \frac{1}{2} (\boldsymbol{R}_g)_{kk} + \mu_k^{(\sigma)} = 0$$

In the interior ($\mu_k^{(\sigma)} = 0$):

$$(\boldsymbol{R}_g)_{kk} = \left[\boldsymbol{u}_{g,k}^\top \hat{\boldsymbol{V}}_g \boldsymbol{u}_{g,k} - (\boldsymbol{\Sigma}_g^{-1})_{kk}\right] = 0$$
$$\implies (\boldsymbol{\Sigma}_g^{-1})_{kk} = \boldsymbol{u}_{g,k}^\top \hat{\boldsymbol{V}}_g \boldsymbol{u}_{g,k} \tag{36}$$

Similarly to the simple model, for each group-component pair (g, k), there is a FOCs of the form $\boldsymbol{u}_{g,k}^\top \hat{\boldsymbol{V}}_g \boldsymbol{u}_{g,k} = \boldsymbol{u}_{g,k}^\top \boldsymbol{e}_k$.

Lastly, analogously to the simple case, the first-order conditions with respect to λ_k for $k = 2, 3$ in the interior ($\gamma^2 > 0$) gives

$$\sum_{g=1}^{G} \boldsymbol{e}_k^\top \boldsymbol{R}_g \boldsymbol{\lambda} = 0 \iff \sum_{g=1}^{G} \boldsymbol{e}_k^\top \boldsymbol{\Sigma}_g^{-1} \hat{\boldsymbol{S}}_g \boldsymbol{w}_g = \sum_{g=1}^{G} \boldsymbol{e}_k^\top \boldsymbol{\Sigma}_g^{-1} \boldsymbol{\lambda} \tag{37}$$

The G equations from (35), the GK equations from (36), and the $K-1$ equations from (37) form a system of $G(K+1)+K+1$ nonlinear equations with $G(K+1)+K+1$ unknowns. Assuming $G = 3$ and $K = 3$, the unknowns are $(\{\gamma_g^2\}_{g=1}^3,\ \lambda_2, \lambda_3, \{\sigma_{g1}^2, \sigma_{g2}^2, \sigma_{g3}^2\}_{g=1}^3)$, which are solved numerically via Newton–Raphson algorithm where the starting points are $\hat{\gamma}_{g,0}^2 = \frac{1}{3}(\hat{V}_{12}^{(g)} + \hat{V}_{13}^{(g)} + \hat{V}_{23}^{(g)})$, $\hat{\lambda}_{2,0} = \frac{\sum_g I_g \left(\frac{\hat{V}_{23}^{(g)}}{\hat{V}_{13}^{(g)}}\right)}{\sum_g I_g}$, $\hat{\lambda}_{2,0} = \frac{\sum_g I_g \left(\frac{\hat{V}_{23}^{(g)}}{\hat{V}_{12}^{(g)}}\right)}{\sum_g I_g}$, and $\hat{\sigma}_{gk,0}^2 = \max\left(0, \hat{V}_{kk}^{(g)} - \hat{\gamma}_{g,0}^2\right)$.

F Censored Exploded Logit - Simulations

This appendix provides details on the Monte Carlo simulation procedure designed to provide evidence of whether the willingness to pay estimates generated with the censored rank-ordered logit model are unbiased.

For each simulation b, I set the number of candidates to $I = 3,000$. To implement the model used in the simulations, I randomly draw 10 job openings from the original dataset ($J = 10$) and for each job posting j that was randomly draw, match the values corresponding to their earnings and many cities dummy variable to w_j and x_j, respectively. Then, I set the true parameters to $\beta_0^w = 2.8$ and $\beta_0^x = 0.3$, normalize the variance of the error term, ε_{ij}, to 1 and its mean to 0 when generating the data for the error term and compute the utility that each individual i assigns to each job opening j. The model of the candidate's i utility obtained from applying to job posting j is:

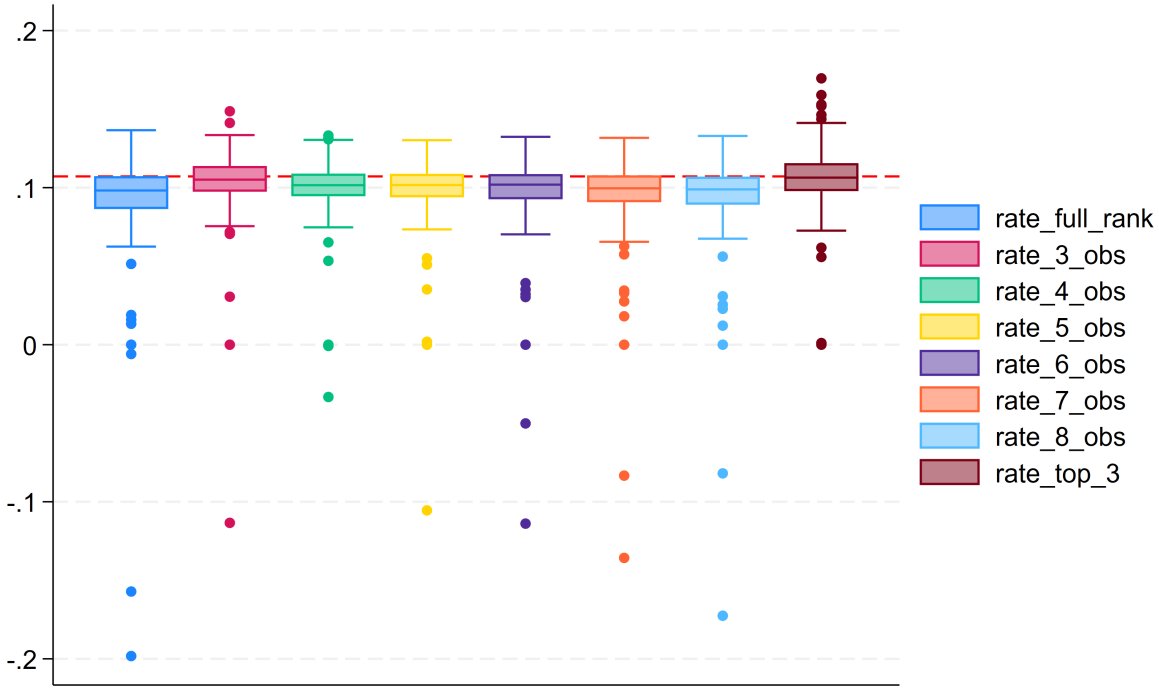
$$u_{ij} = \beta^w \ln w_j + \beta^x x_j + \varepsilon_{ij},\ \varepsilon_{ij} \overset{iid}{\sim} EVT1 \tag{38}$$

After computing the numerical value of the utility that each individual obtains from each job posting j, construct a complete rank-ordered list for each individual i based on their simulated utility values for all the randomly draw job openings. Subsequently, randomly draw, for each applicant, $R = 3, ..., J-1$ job openings, to mimic the censored nature of of the rank-ordered lists in the original dataset. Then, for each value of R, construct a short rank ordered list for each individual i

based on the simulated utility values of their randomly draw job openings and run a rank-ordered logit for the full list and a censored rank-ordered logit for $R = 3, ..., J-1$. Lastly, after estimating $\hat{\beta}^w$ and $\hat{\beta}^x$ for each list size, compute the rate $\hat{\beta}^x/\hat{\beta}^w$ and the WTP $1 - exp(-\hat{\beta}^x/\hat{\beta}^w)$. Given our initial true parameters, the true value of the rate is $0.3/2.8 = 0.1071428571$ and the true value of the WTP is 0.101602.

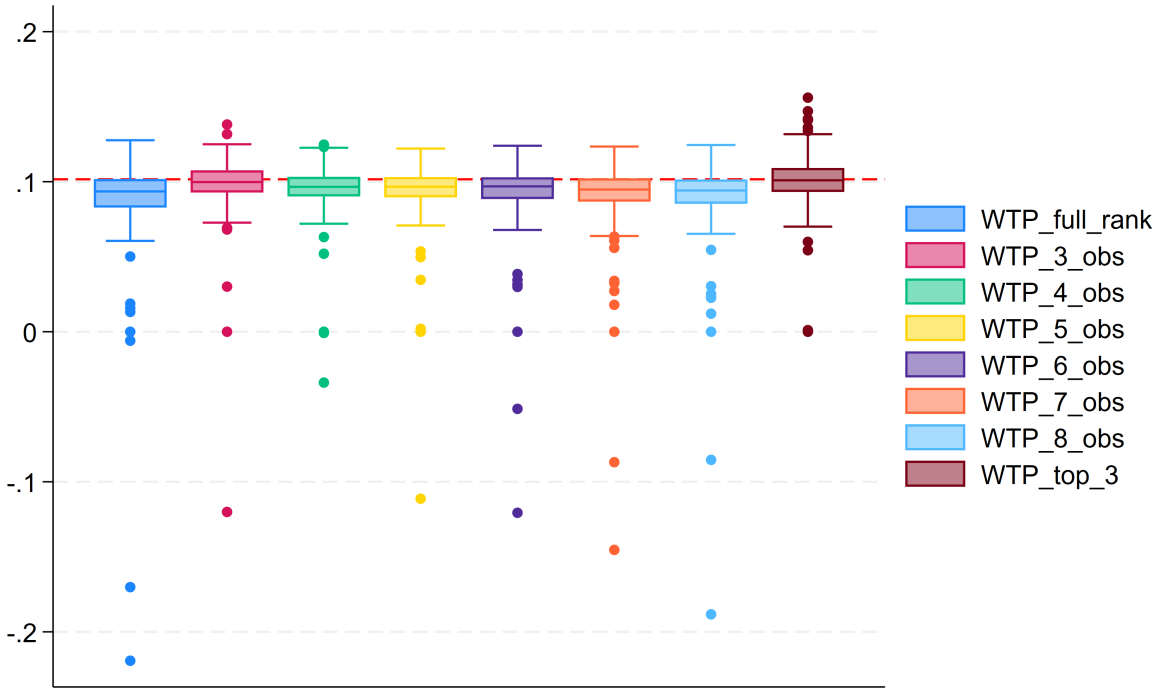
In the simulations presented below, I set $B = 300$ as the total number of samples.

Figure 5: Distribution of the simulated $\hat{\beta}^x/\hat{\beta}^w$, $B = 300$



Notes: The graph shows the box-plots of the simulations of the rank-ordered logit for the full list and of the censored rank-ordered logit for $R = 3, ..., J-1$ and using only the top 3 simulated choices of each candidate. The value indicated by the dashed line corresponds to the value of the rate of the two true parameters, $\beta^x/\beta^w = 0.3/2.8 = 0.1071428571$.

Figure 6: Distribution of the simulated $1 - exp(-\hat{\beta}^x/\hat{\beta}^w)$, $B = 300$



Notes: The graph shows the box-plots of the simulations of the rank-ordered logit for the full list and of the censored rank-ordered logit for $R = 3, ..., J - 1$ and using only the top 3 simulated choices of each candidate. The value indicated by the dashed line corresponds to the value of the willingness to pay for the attribute "many cities" given the true parameters, $1 - exp(-\beta^x/\beta^w) = 1 - exp(-0.3/2.8) = 0.101602$.